

Allometric calibration bias in tropical forest carbon

Architecture-informed measurement, LiDAR dependence and annual greenhouse-gas attribution

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Abstract

Forest-carbon estimation links a biological reference, a spatial prediction and an annual carbon account. We assemble these operations into a framework that separates accuracy of the reference from precision of its spatial reproduction. Conditional-expectation identities show that an ideal LiDAR predictor retains the conditional bias of its training labels; a population-transfer identity isolates the effect of changing architectural composition. Stock-weighted covariance determines when a carbon-stock correction transfers unchanged into annual release. The paired-account and finite-increment results distinguish common-scale cancellation from differential amplification of sources and removals. A published-digit analysis resolves two class assignments and characterizes a shared normalization in the initial 723.97 PgC arXiv application. A stem–branch volume decomposition turns its high-density class into an explicit test of joint basal area, height, material density and organ allocation. Illustrative low-cylinder scenarios fall below that target, whereas the cited mean crown allocation does not establish a class-wide physical ceiling. Three conditional accounting outcomes connect a supported forest-source correction to a larger total, increased forest attribution within a retained total, or joint source–removal revision. A matched historical scenario demonstrates a largest-land-group ranking with other absolute contributions retained; independent constraints govern empirical application. These results direct calibration towards independently measured organ volumes and tissue carbon, representative spatial weights and disturbance-specific observations. The framework identifies what must change in an estimator and what must change in an annual account, while keeping empirical support for the allometric-bias mechanism distinct from validation of the initial map’s magnitude.

Keywords: allometric calibration bias; tropical forest carbon; tree architecture; lidar; reference-label bias; biological volume; tissue carbon; deforestation and degradation; stock-to-flux transfer.

1 Introduction

The reliability of forest-carbon accounting begins with the biological quantity assigned to trees and stands, not with the spatial precision of the map used to distribute it. Allometry-based carbon labels may omit or misrepresent architecture and organ-level composition even when tree position, diameter or canopy height is accurately measured. When those labels calibrate a remote-sensing product, increased coverage can reproduce the same carbon-scale error over a larger area. A sensor measures properties of the forest; the conversion of those properties into biomass and carbon remains a separate inference.

This article examines the propagation of structural and compositional bias in allometry-based forest-carbon estimation into spatial products and annual greenhouse-gas accounts. Architecture-informed measurement and the initial pantropical reassessment provide the methodological basis and application

[1, 2]. The later NEBIOT essay discusses their interpretive implications [3]. Their central mechanism is that the actual material of stems, branches and other organs can differ from the material represented by a generalized prediction. Measuring that architecture is consequently a way to diagnose allometric bias, rather than an unrelated explanation for the resulting discrepancy.

The contribution is the assembly of the measurement, calibration and accounting operators. The same weighting principle exposes reference-label inheritance, stock-to-flux covariance and different source, removal, gain and loss corrections; the relevant weights and biological quantities change at each stage. The classical identities are used to specify observable mechanisms and discriminating tests. Source-table diagnostics and detailed material scenarios support that argument in the technical supplement.

Status of the initial estimate. The 723.97 PgC value is the initial architecture-informed estimate in arXiv:1508.03667, first posted on 14 August 2015 and revised on 17 June 2016; it is a preprint result, not a peer-reviewed inventory determination. It covers the mapped pantropical aboveground-vegetation domain excluding the Australian tropics, with nominal map epoch 2007–2008 [1, pp. 23–24]. The author is developing successor calibrations; this article supplies neither a replacement map nor a newly validated total. Its analytical results are independent of that initial numerical value. Independent empirical verification of the class-5 mean remains outstanding.

The evidence synthesis is purposive rather than a meta-analysis. The author developed the architecture-informed methodology underlying the cited preprint. Original measurements, source-reported method comparisons and analytical identities are identified separately; no synthetic stand is represented as a new field observation. Figures 1–3 connect the reported inventory, the forest-carbon interpretation and the two calibration pathways. Figure 2 presents the numerical second interpretation with a questioned total, a double-headed arrow and a conditional largest-sector comparison, while keeping the standing-stock record separate from annual flows. The sector–activity–gas layouts of the first two figures adapt the WRI and Ecofys visual convention [4, 5]; intermediate activity nodes are conceptual pathways, not an alternative official sector classification.

2 Allometric calibration bias as a measurement problem

2.1 Architecture as a mechanism of allometric calibration bias

Aboveground biomass allometry predicts dry mass from a selected set of tree attributes. A representative formulation is $\widehat{B}_A = g_\theta(D, H, \rho)$, where diameter, height and density provide informative but incomplete descriptions of a tree. An architecture-informed estimate instead resolves the material in its organs. Two individuals sharing the predictors of a reduced-input relationship can differ in crown extent, branch hierarchy, buttress form, damage and allocation among organs. A population-average prediction need not recover either individual’s mass exactly. Systematic differences between the calibration population and the forest being assessed can then turn unresolved individual variation into aggregate bias.

Underrepresented branch or stem material is a physical mechanism of allometric underestimation. A systematic discrepancy can therefore arise within the woody material already assigned to an aboveground-biomass target. The relevant comparison is between predicted and actual material in the same woody and nonwoody compartments. This distinction also separates target completeness from estimator accuracy: an equation may be intended to predict total aboveground biomass and still represent the distribution of that biomass imperfectly.

Goodman et al.’s 51-tree sample provides direct evidence of the architectural mechanism: mean crown allocation was $44\% \pm 2\%$ of aboveground biomass, with individual fractions of 14–71%, and the tested height-inclusive pantropical equations underestimated the sample total by 11–14% [6]. Their crown component includes branches and attached foliage/reproductive material; ten stems used sectional volume–density estimation rather than complete weighing. Crown measurements improved prediction. A

broadly, partly overlapping synthesis of 673 harvested trees found crown fractions of approximately 3–88% and a size-dependent allocation pattern [7]. These are empirical distributions and estimator comparisons, not universal limits on crown volume or class-average carbon.

2.2 Conditional representation and transfer between populations

Let X contain the recorded predictors and let Z contain additional architecture and composition. Write true biomass as a square-integrable function $B = \phi(X, Z)$ on the defined pool. For an ideal squared-error predictor calibrated in population P , $g_P(x) = \mathbb{E}_P[B \mid X = x]$. Even this ideal predictor has residual variation

$$\mathbb{E}_P[(B - g_P(X))^2] = \mathbb{E}_P[\text{Var}_P(B \mid X)]. \quad (1)$$

The conditional-mean property gives zero mean prediction error in P ; unresolved variation alone is therefore not a proof of population bias. Its role changes when the conditional distribution of architecture or composition changes. For a target population Q on overlapping predictor support,

$$g_P(x) - \mathbb{E}_Q[B \mid X = x] = \int \phi(x, z)\{dP(z \mid x) - dQ(z \mid x)\}. \quad (2)$$

The expression is exact when the same biological response function is used and the expectations exist. Finite-sample fitting, predictor measurement errors and a misspecified functional form add further errors. The formula identifies a mechanism: using the same diameter and height range does not establish that the unrecorded architecture is distributed in the same way. A downward error requires the conditional biomass in the target population to exceed the calibration expectation, rather than being assumed from heterogeneity alone.

A synthetic example illustrates the distinction. Suppose trees share the recorded predictors but have two architectural states with biomass 100 and 200 arbitrary units. Equal calibration frequencies give a prediction of 150. If the target frequencies are 0.2 and 0.8, the target mean is 180 and the transferred prediction is 30 units lower, a 16.7% shortfall. Increasing the number of trees assessed with the unchanged rule does not alter that expected discrepancy. These are analytical values, not measured tropical forest corrections; the executable example is provided in the numerical companion.

Homogeneous species, ages and management can narrow the conditional variation that a fitted model must represent, but homogeneity is not the criterion defining the validity of allometry. Chave et al. evaluated equations against 4,004 harvested trees from 58 tropical sites and reported broad predictive performance [8]. Applicability is therefore established by validated transfer across architecture and composition, rather than by a crop-versus-natural-forest classification. Crop and plantation relationships require calibration checks, while natural-forest relationships can be useful within independently supported domains. Treating either as an independently measured carbon scale without testing that transfer is the methodological error addressed here. What heterogeneous tropical forests particularly demand is explicit testing of architecture, composition and disturbance states that reduced predictors may fail to distinguish.

2.3 The working hypothesis and its observable consequences

The central hypothesis is that structural and compositional underrepresentation in the allometric calibration of some tropical forest populations produces consequential errors in carbon stocks and the fluxes derived from them. The architecture-informed reassessment supplies the methodological basis and application [1, 2]. Its reported discrepancies motivate measurement at the level where the error is proposed to arise: organs, trees and vegetation populations, followed by the spatial and temporal operations used in accounting.

The hypothesis has observable consequences. At matched tree and plot supports, independent reconstructed material and tissue carbon should explain signed residuals of the allometric estimator, with

residual structure related to architectural or compositional attributes omitted from that estimator. Replacing biased reference labels should alter remote-sensing carbon predictions even when the sensor observations are unchanged. Disturbance-weighted residuals should then predict the correction to the source population more directly than the unweighted mean correction across intact forest. These predictions admit localized support, mixed signs or failure; their measurement does not require treating conventional inventory agreement as the definition of correctness.

3 Why LiDAR coverage does not by itself resolve carbon-scale bias

3.1 Sensor observations, calibration labels and independent reference

LiDAR supplies geometric information through laser ranging, return waveforms or point clouds. Biomass and carbon require additional material or empirical relationships. The crucial distinction is therefore the *inference pathway*, not the sensor name. A height-based biomass regression calibrated on allometric plot estimates remains dependent on those reference estimates. Detailed laser reconstruction of material volumes combined with independently measured density and carbon fractions can instead reduce that dependence. Both pathways have uncertainties, but they are not the same measurement procedure.

The documented GEDI Level-4A Version-2 pathway relates waveform relative-height metrics to field-plot estimates of aboveground biomass density, using regional and plant-functional-type models. Its guide describes geographic cross-validation and flags observations outside predictor or response support [9]. These controls address transfer and prediction errors; validation against reference labels alone cannot expose a systematic error common to the labels. This observation is not an estimate of the bias of a particular GEDI release. It identifies a component of reference uncertainty that must be established from the underlying field methods.

Conversely, Calders et al. reconstructed woody volume from terrestrial laser scanning and combined it with basic density in an Australian eucalypt forest. Against 65 harvested reference trees, the tested allometric totals were 29.85–36.57% low, whereas the TLS reconstruction was 9.68% high [10]. This demonstrates a route through direct geometric reconstruction rather than the same biomass allometry; it also demonstrates that laser-derived reconstruction is not error-free. Registration, occlusion, segmentation, branch geometry, leaf–wood separation and material-property sampling require their own checks. Thus increased sensing coverage cannot cure a biased calibration merely by being extensive, while appropriately used LiDAR can be part of the architectural solution.

Table 1: Three distinct measurement pathways. The same sensor can participate in different calibration chains; independence is an attribute of the chain, not of spatial resolution alone.

Pathway	Carbon information	Principal calibration question
Field predictors plus biomass allometry	Empirical mass prediction from recorded attributes	Does calibration represent the target architecture, composition and disturbance?
Remote structure plus allometric reference labels	Spatial prediction of the reference biomass/carbon scale	Does validation test true material, or repeat the label-generating estimator?
Organ reconstruction plus material analysis	Volume, basic density and tissue carbon, then spatial extension	Are the volumes and properties independently measured and spatially representative?

3.2 Segmented terrestrial reconstruction and plot-scale processing

Disney et al. describe how terrestrial laser scanning (TLS) resolves tree form and woody material independently of a conventional biomass scaling equation [11]. The relevant sequence is registration and

leaf–wood classification, individual-tree segmentation, quantitative structure modelling, and conversion of reconstructed volume using material properties. Segmentation alone supplies neither biomass nor elemental carbon. This pathway can supply new reference measurements, rather than merely predict the allometric labels used by an existing map; geometry, density and chemistry nevertheless retain separate uncertainties.

A direct regional example is Calders et al.’s Wytham Woods study, coauthored by Disney [12]. Across 1.4 ha and 815 live trees, TLS-based aboveground biomass was 409.9 Mg ha^{-1} , 1.77 times the estimate from the comparison allometries. This was a temperate stand reconstruction using quantitative structure models and assigned species-density information, not complete independent mass weighing. It supports the possibility of substantial reference-calibration bias and a route to reducing it; it does not establish a pantropical multiplier or validate 723.97 PgC .

Wilkes, Disney and colleagues’ TLS2trees pipeline addresses automated, plot-scale segmentation [13]. Its 7.5-ha, ten-plot demonstration covered seven forest types and segmented 10,557 trees; 1,053 were paired with manually segmented reference trees. Their Table 2 reports instance-segmentation wall times of 16 and 18 h for the one-hectare NOU and MLA datasets, with maximum RAM per tile of 51.6 and 129.8 GB, respectively. These are implementation-specific benchmarks, not universal per-hectare costs. They substantiate the computational bottleneck while showing why tiled, parallel processing is useful. Errors in tree separation and small-tree recovery remain relevant to complete plot biomass. Thus scalable sensing, scalable segmentation and independently accurate material calibration are three distinct requirements.

3.3 Calibration-label inheritance

Let C denote true carbon on a fixed pool and support, $Y = C + b + \varepsilon$ the calibration label, and L the remote-sensing predictors. Here b is the label’s signed systematic discrepancy and $\mathbb{E}_P[\varepsilon | L] = 0$. In the population limit of squared-error prediction, define $m_Y(l) = \mathbb{E}_P[Y | L = l]$ and $m_C(l) = \mathbb{E}_P[C | L = l]$. Linearity of conditional expectation gives

$$m_Y(l) - m_C(l) = \mathbb{E}_P[b | L = l]. \quad (3)$$

This is an exact conditional-mean identity, not a claim of a new statistical theorem or of universal consistency for a particular learning algorithm. If a predictor accurately reproduces Y , its discrepancy from C can still equal the conditional label bias. An average-zero label error independent of L behaves differently: it contributes noise but no conditional-mean displacement in this ideal case. A target-population shift adds a second component,

$$m_Y(l) - \mathbb{E}_Q[C | L = l] = \mathbb{E}_P[b | L = l] + \{\mathbb{E}_P[C | L = l] - \mathbb{E}_Q[C | L = l]\}. \quad (4)$$

Separate reference-mass validation and transfer diagnostics are consequently required. Where predictors cannot distinguish structurally different trees or plots, accurate ranging cannot supply the omitted material information through a response label that already misrepresents it.

A complementary shared-error example makes the coverage distinction explicit. If $Y_i = C_i + b_0 + \varepsilon_i$, with a fixed shared offset b_0 and independent zero-mean errors of variance σ^2 , then the mean error has mean-squared value

$$\mathbb{E}[(\bar{Y}_n - \bar{C}_n)^2] = b_0^2 + \sigma^2/n. \quad (5)$$

More samples reduce the independent component, not the squared shared offset. Correlated errors require their full covariance instead of σ^2/n . Actual wall-to-wall pixels are not independent reference measurements, and a denser point cloud should not be equated with n independent carbon calibrations. The result separates spatial detail, statistical precision and absolute carbon accuracy without denying the value of better sensing.

3.4 Replacing the reference rather than merely refining the map

The constructive response is to calibrate the carbon scale from independently checked architecture and tissue properties, while using remote sensing to extend those labels through a documented support domain. Randomly splitting outputs from one label-generating model only tests prediction of that model. Complete site and vegetation holdouts, independent material references, and signed residuals against both allometric and architecture-based estimates provide stronger evidence. The same safeguard applies when an architectural model supplies training labels: geometry must be validated before label replication is interpreted as physical accuracy.

Calibration improvements may change the regional or pantropical point estimate without invalidating the measurement argument. The 723.97 PgC estimate reported on arXiv remains the reported application; it is neither recalculated from a LiDAR product here nor made the value that future calibration must reproduce. The methodological objective is the fidelity of the material-to-carbon chain and the consequences of replacing biased reference quantities. Figure 3 summarizes the two routes from field information to spatial products and annual accounts.

4 Architecture-informed measurement and the pantropical application

4.1 The architectural measurement chain and its physical tests

A dimensionally explicit tree-carbon representation is

$$C_j = \sum_o V_{jo} \rho_{jo}^b f_{jo}^C, \quad (6)$$

where non-overlapping compartments o have green material volume V , basic density ρ^b defined as oven-dry mass per green volume, and dry-mass carbon fraction f^C . With consistent units, the product is carbon mass. Hollow space, bark, leaves and branches must not be double counted; roots do not belong in an aboveground-only target. Moisture conventions must be matched rather than multiplying hydrated mass by a dry-mass carbon fraction.

Pantropical allometries can incorporate diameter, height and wood density [8], yet trees sharing those predictors can differ in crown architecture, branch allocation, hollow volume and tissue composition. A relationship based on the reduced predictors can describe their calibration-population average without recovering every individual's material distribution. Architecture-informed reconstruction resolves additional quantities, so its validity is not conditional on reproducing an allometric prediction. Tissue chemistry provides a separate measurement axis: a biomass correction and a carbon-fraction correction may have different signs [14]. The IPCC tropical aboveground default of 0.47 is an accounting parameter for specified applications, not a substitute for measured organ fractions [15].

NEB002 combines measured architecture, numerical integration or voxelization, material density and elemental carbon measurements; NEB001 links vegetation identity and environmental structure to spatial mapping [2]. Neural networks extend the resulting carbon estimates. The physical inputs therefore precede the statistical extension. Validation is assessed at each step: representation of measured material, assignment of density and chemistry, prediction beyond the reconstructed individuals, and integration across vegetation types. A model's ability to reproduce its training targets is relevant to the prediction step but does not summarize all the evidence supporting those targets.

The preprint reports 221 reconstructed individuals from 127 species (p. 6), and 2,232 tissue samples from 186 individuals processed with a LECO CHN-600 analyzer (pp. 16–17). Chemical quality control used reference compounds at ten-sample intervals. Its physical volume comparison used 10 individuals of *Aphelandra pulcherrima*, 1.54–2.9 m tall, with cutting, weighing and water displacement (p. 25 and Fig. 11). The comparison is summarized in Table 2; the measured material and the geometrical predictions

are distinct observations [1].

Table 2: Reported physical-volume validation in the original preprint [1, p. 25]. The same 10 *Aphelandra pulcherrima* individuals provide the reference and reconstruction comparison. Mean volumes are source-reported; the final column gives the non-negative shortfall from the water-displacement reference, calculated from the displayed means and rounded to one decimal place. All three reconstructions fall below the reference, with the 1-mm voxel grid showing the largest shortfall. It is a difference of sample means, not an individual-error bound or a pantropical confidence interval.

Method	Mean volume (cm ³)	Shortfall from reference (%)
Water displacement (reference)	1,240.00	0.0
Numerical integration	1,217.76	1.8
Voxelization, 1-cm resolution	1,211.00	2.3
Voxelization, 1-mm resolution	1,189.92	4.0

This is physical validation of volume at the tested scale, alongside laboratory measurements of tissue carbon; it cannot be reduced to neural-network self-consistency. Its extension to different sizes and architectures is a distinct validation question. Numerical accuracy in integrating a specified mesh and accuracy of the mesh as a representation of a tree also remain separate quantities. Independent geometry, density and chemistry checks address these steps without requiring agreement with a conventional biomass equation.

The regional chain likewise precedes the pantropical total. The preprint reports network-based carbon estimates for more than 6,000 individuals in 27 vegetation types and a wider regional comparison across 54 vegetation types. Their reported average carbon densities were 177.03 MgC ha⁻¹ for the architecture-informed method and 61.41 for the allometric comparison. The pantropical calibration used regional carbon targets with the carbon map published by Baccini et al. in 2012 and the height map published by Simard et al. in 2011 [1, pp. 22–24]. Predictor and calibration dates are distinguished in Table 5. A conventional map used as a predictor is not thereby accepted as the true carbon target; shared predictors can nevertheless induce dependence between products. The complete chain is measured organs → reconstructed trees → vegetation-specific carbon estimates → spatial calibration, rather than a post hoc multiplication of an accepted global inventory.

4.2 Calibration support and material requirements

The reported regional mean of 177.03 MgC ha⁻¹ is not the largest calibration response. Page 23 of the arXiv source states predictor ranges of 0–190 MgC ha⁻¹ for the Baccini map and 0–38 m for Simard height, with regional carbon responses spanning 0–523 MgC ha⁻¹ [1]. The class-5 lower edge, 478, is within that marginal response range; its mean and maximum exceed the endpoint. Thus some predictions extend beyond the recorded response range, but the mean’s exceedance does not place every class-5 cell outside it or establish extrapolation in the multivariate predictor domain. Joint predictor support and signed error require record-level labels, model inputs and independently held-out observations.

If the 478–605 MgC ha⁻¹ interval is the implemented raster definition and the ratio 536.33 is its area-weighted mean on the stated mask, at least 16.25% of class-5 area (86.4 million hectares) exceeds the page-23 response endpoint of 523. Capping only that exceedance at 523 removes between 7.08 and 20.01 PgC, or less than 2.77% of the reported pantropical total. The printed-mean convention gives a minimum bound of approximately 9.82% (52.2 million hectares) and a 4.28–18.20 PgC clipping change on the same illustrative area. Supplementary Note S1 derives the sharp moment bounds and shows that the affected area can nevertheless be as large as the whole class. These bounds concern a specified numerical cap, not the magnitude of calibration error: biased in-range labels, joint predictor extrapolation and errors in other classes can persist after capping. Page 22 separately reports a regional value of 523.4 MgC ha⁻¹;

the archive supplies this nearby-endpoint sensitivity rather than treating the rounded page-23 range as a recovered exact label maximum [1].

If every mapped compartment's carbon fraction lies in the preprint's measured tissue range 0.4111–0.4999, the class edge implies at least about 956 Mg dry biomass ha⁻¹, and the ratio-based class mean implies about 1,073–1,305 Mg ha⁻¹. These are conditional bounds from the reported tissue range, without choosing 0.47 as a uniform fraction; they are not a reconstructed biomass map or a demonstrated global tissue-fraction bound.

The initial map's standing and successor-calibration programme are specified in the [status box](#). Alternative totals also depend on their domains: Feldpausch et al.'s 285.1 PgC exceeds Baccini's 228.7 on a different mask, while representing a downward height correction from 320.5 on Feldpausch's own unchanged mask; Avitabile et al.'s fusion also lowers its input estimates [16–18].

Interpreting the high-density requirement. At the illustrative carbon fraction 0.47 and basic density 0.60 Mg m⁻³, the ratio-based class mean requires approximately 1,900 m³ ha⁻¹ of wood when the separate remainder is zero. A stem–branch decomposition in Supplementary Note S1 shows that a scenario with stem form ratio 0.4–0.6, branch fraction 0.44 and equal compartment densities gives whole-wood ratios of 0.71–1.07. Combined with basal area 30–40 m² ha⁻¹ and weighted height 30 m, it yields approximately 181–363 MgC ha⁻¹, below the class target. At basal area 50 and weighted height 40, however, the same parameter range gives approximately 403–604 MgC ha⁻¹, spanning the target. These are sensitivity configurations, not observed class means or established plot maxima. The published crown fractions are sample distributions; their mean cannot serve as an upper bound on the required cylinder-weighted volume ratio. The comparison identifies the joint structural conditions needed to attain the mapped density, whose class-average empirical support remains unverified. Independent joint upper bounds could reject the class mean even without the original raster; the cited studies do not establish such a bound for its population. Representative geometry, material and sampling records remain necessary to validate that mean.

Acquisition and discretization provenance. The preprint describes field dimensions, photographs, scientific drawings and procedural reconstruction with SpeedTree and Xfrog [1, pp. 6–10]. The accessible record does not assign each of the 221 trees to a mutually exclusive capture route or identify an independent geometric check for every tree. Modern acquisition options cannot be assigned retrospectively to those records. The companion capture-route schema separates observed inputs, procedurally generated organs, volume algorithm, mesh identifiers and independent checks; its row set is deliberately empty until primary records are supplied.

The 1-mm voxel mean falls farther from both numerical integration and displacement than the 1-cm mean. A non-watertight mesh combined with an inside/outside rule is one testable cause; changed meshes, scale or origin, different occupied-cell rules and averaging also require checking. A consistent sequence on one closed mesh should converge as the grid is refined, although successive errors need not decrease monotonically. The aggregate means do not identify the cause; the original mesh and paired inclusion settings distinguish these mechanisms.

4.3 Biological material is not merchantable timber

The physical target is the material in the defined biological compartments, irrespective of saleability. For disjoint woody components on the same live aboveground boundary,

$$V_W = V_{\text{merchantable}} + V_{\text{nonmerchantable}}. \quad (7)$$

Neither a minimum commercial top diameter nor timber quality removes existing carbon. Stem tops, branches, twigs, buttresses and noncommercial individuals must enter the target where present, with

foliage or other nonwoody organs treated separately. Cavities contain no woody material; carbon remaining in degraded tissue must be represented with appropriate material properties and live/dead assignment rather than being excluded for lack of timber value. An outer crown envelope contains air and is not a material volume.

IPCC Box 4.2 explicitly distinguishes merchantable volume from total biomass and describes expansion of timber-volume records to additional components [15]. This is a definition of the measurement boundary, not validation of any default expansion factor. Direct biomass allometries may also target whole-tree aboveground dry mass, so it would be incorrect to equate every allometric output with saleable timber. The structural issue is whether the selected predictor or expansion actually represents the nonmerchantable material of the target vegetation.

For measured volume V and basic density ρ^b defined as oven-dry mass per corresponding green volume, $V\rho^b$ estimates dry mass; multiplying by the matched dry-mass carbon fraction gives carbon. Hydrated mass, green volume, dry volume and elemental-carbon fraction cannot be interchanged. Density and carbon fractions vary by organ and sample, and their products must be aggregated with the relevant material weights. A single assumed fraction is useful for an explicitly marked sensitivity calculation, not as a retroactive description of a method that measured tissue carbon directly.

Consequently, a hypothetical large unmeasured foliage remainder is not required to explain underestimated woody biomass. The disagreement can concern the woody material itself. Component completeness, prediction error within already included components, and spatial selection are distinct diagnostic questions. None should be used to replace the allometric-bias mechanism with an unsupported assumption that the conventional woody subtotal is exact.

4.4 Intermediate stems, abundance and forest degradation

Forest-carbon estimation and release accounting concern the full sampled size distribution. The architectural preprint compares organ biomass across diameters of 1–20 cm in Figure 6, extends its comparison to larger diameters in Figure 7, and presents carbon estimates for more than 6,000 individuals in Figure 8 [1, pp. 20–22, Figs. 6–8]. Its Table 1 contains, for example, aboveground-biomass estimates of 104.82 kg for *Inga* sp. 10 (15.24 cm DBH), 109.84 kg for *Pouteria* sp. 4 (11.58 cm DBH), and 115.74 kg for *Virola elongata* (14.63 cm DBH). These are biomass outputs, whereas Figure 8 uses kilograms of carbon. The role of intermediate stems is therefore part of the original measurement domain, rather than an extrapolation from a large-tree validation sample.

A complementary stand-structure result appears in Vásquez and Arellano [19, Table 170, p. 942]. In the *Acalypha* sp.–*Guazuma ulmifolia* community, the 10–30 cm class contributes a reported 38.97% of aboveground biomass and 38.79% of carbon. Dividing its 2.88 tonnes of biomass by 30 stems per 0.05 ha gives 96 kg biomass per stem (Table 3). This ratio of displayed class aggregates illustrates an approximately 100 kg mean, not an assertion that every stem has that mass. Classes II and III together account for 62.71% of that community’s biomass. “Intermediate” here follows the local diameter classes; it is not a universal mass threshold.

The 2012 chapter estimated biomass using selected Chave et al. relationships and measured tissue-carbon fractions; the later architectural reconstruction is a distinct method. Its *Acalypha*–*Guazuma* community total is 7.39 Mg per 0.05 ha, or 147.8 Mg ha⁻¹, placing the abundance example in a comparatively low-biomass stand within that study [19, Table 170, p. 942]. It illustrates a size-class distribution, rather than supplying independent support for the class-5 density. The near-100 kg architectural records instead establish the size coverage of that method. Together with stands where larger classes dominate, the examples motivate vegetation-specific population weighting.

Table 3: A published intermediate-stem example: the *Acalypha* sp.–*Guazuma ulmifolia* community in Córdoba [19, Table 170, p. 942]. Class II is 10–30 cm DBH in the table. Inputs and percentage shares are source-reported; means per stem are derived from the displayed aggregates. The chapter’s abstract gives a different lower-class interval, so the table’s explicit bin definition is used here.

Quantity	Value	Basis
Stems per 0.05 ha	30	Reported
Biomass (tonnes per 0.05 ha)	2.88	Reported
Carbon (tonnes C per 0.05 ha)	1.30	Reported
Share of community biomass (%)	38.97	Reported
Share of community carbon (%)	38.79	Reported
Mean biomass per stem (kg)	96.00	Derived
Mean carbon per stem (kgC)	43.33	Derived
Classes II + III biomass share (%)	62.71	Reported

For disjoint classes h , let n_h be stem abundance and C_j the carbon of an individual. Then

$$C_h = \sum_{j \in h} C_j = n_h \bar{C}_h, \quad F_h(t) = \frac{44}{12} \sum_{\tau \leq t} \sum_{j \in \mathcal{D}_{h,\tau}} \sum_o C_{jo,\tau} d_{jo,\tau} k_{jo,\tau}(t - \tau).$$

Here $\mathcal{D}_{h,\tau}$ indexes affected stems, o indexes disjoint organs, $0 \leq d_{jo,\tau} \leq 1$ is the affected fraction, and k is the release rate as CO_2 in yr^{-1} . Thus stocks in kgC give release in $\text{kgCO}_2 \text{ yr}^{-1}$. Repeated disturbances must use the remaining carbon and disjoint affected cohorts. Total forest release is $F = \sum_h F_h$: per-tree mass, abundance, exposure and release history all enter. A class containing lighter but numerous affected stems can supply more carbon or annual release than a sparse large-tree class. Partial crown and branch losses, collateral mortality, understory removal and repeated damage belong in degradation accounting even when land remains classified as forest. Regrowth and changes in future uptake are separate terms. These definitions are consistent with the cohort formulation in Section 6 and degradation-process distinctions in Lapola et al. [20].

5 Spatial, temporal and accounting boundaries

5.1 Stocks, transfers and gas-equivalent quantities

Aboveground biomass is dry biological mass; aboveground carbon is the carbon contained in that mass. Although $1 \text{ Pg} = 1 \text{ Gt}$, a stock in PgC cannot be inserted into a flow in GtC yr^{-1} . For carbon actually emitted as CO_2 , the conversion is

$$E_{\text{CO}_2} = (44/12)E_C. \quad (8)$$

It preserves, rather than creates, the time dimension. Movement from live wood to dead wood or harvested products is not necessarily an atmospheric emission in the same year. Carbon pools, transfers and eventual release therefore require explicit representation [15].

A multi-gas inventory instead applies a selected climate-equivalence convention,

$$T_{\text{GHG}} = \sum_g \text{GWP}_{100,g} E_g. \quad (9)$$

This is not a conserved carbon-mass balance. A CO_2 component can legitimately form the numerator of a CO_2e share because the equivalence factor for CO_2 is one; the denominator must still specify the other gases, time horizon and accounting boundary. Gas conversion does not make a net land-use category equivalent to gross tropical deforestation.

Throughout, a *gross component* means an explicitly positive emission or removal within the selected account. It does not mean that an anthropogenic inventory includes all ecosystem respiration or gross primary production. Those ecosystem fluxes cannot be added indiscriminately to an inventory of human-induced emissions.

5.2 The contemporary reference

Table 4: Reported reference quantities through 2025. Values are rounded source estimates or projections, not recalibrated forest-carbon results. LULUCF denotes land use, land-use change and forestry.

Quantity	Annual value		Boundary and status
EDGAR GHG excluding LULUCF	54.1 GtCO ₂ e		2025 inventory; AR5 GWP ₁₀₀
EDGAR GHG including net LULUCF	Approximately 55.0 GtCO ₂ e		Same reported account including its land balance
EDGAR Forest Land	−5.5 GtCO ₂		Living-biomass net removal; remaining forest and land converted to forest
EDGAR Deforestation	+3.7 GtCO ₂		Net category, all regions; includes tropical forest-fire CO ₂
EDGAR Fires	Approximately 2.2 GtCO ₂ e		Allocated fire account; tropical forest-fire CO ₂ excluded to avoid overlap
EDGAR Organic Soils	About +1.1 GtCO ₂		Reported CO ₂ subtotal; the category also includes N ₂ O
EDGAR Other	See signed balance		Non-biomass forest pools and remaining land categories; not independently extracted here
EDGAR net LULUCF	About +0.9 GtCO ₂ e		Sum across the five categories above
GCB net land-use CO ₂	About +4.1 GtCO ₂		2025 projection, low confidence; different attribution convention

The booklet’s Section 3, Figure 6 and Annex 2 define these entries [21]. Forest Land includes land converted to forest; tropical forest-fire CO₂ belongs to Deforestation, with non-CO₂ fire gases assigned separately. The organic-soil number is a CO₂ subtotal. The rounded closure residual, $0.9 - [3.7 - 5.5 + 2.2 + 1.1] \approx -0.6 \text{ GtCO}_2\text{e yr}^{-1}$, consequently absorbs missing subcomponents and rounding as well as Other. The non-land gas composition is 73.8% fossil CO₂, 17.6% CH₄, 5.3% N₂O and 3.3% F-gases; power and heat contribute nearly 30%. The globally implemented Tier 1 Gain/Loss forest account combines land-change products with default growth, expansion, density and allocation factors, including inputs predating the inventory year.

EDGAR’s net +0.9 GtCO₂e and GCB’s +4.1 GtCO₂ projection refer to different gases, land definitions and attribution rules [21, 22]. A primary harmonization study makes the definitional effect explicit: for 2000–2020, Grassi et al.’s bookkeeping-model +4.8 and national-inventory -1.9 GtCO₂ yr^{−1} differ by 6.7; adding the model-estimated managed-forest environmental sink of -6.4 reduces that gap to 0.3 [23]. This demonstrates a consequential reporting difference, separate from the physical calibration of tree mass.

5.3 Pantropical coverage, worldwide extension and map dates

The initial map’s extent and epoch are defined in the [status box](#). Baccini’s predictor covers woody vegetation beyond forest-only formations; a global height predictor does not extend the carbon map’s geographical support [16].

Let P be the actual mapped pantropical domain and $W \supseteq P$ a harmonized worldwide aboveground-

vegetation domain. For the same carbon pool and reference epoch t_0 , spatial additivity gives

$$C_W^{\text{AG}}(t_0) = C_P^{\text{AG}}(t_0) + C_{WP}^{\text{AG}}(t_0).$$

Retaining 723.97 PgC for $C_P^{\text{AG}}(t_0)$, a complete estimate on this wider support includes additional positive stock outside P , including the Australian tropics and temperate and boreal regions. Its increment is not quantified by the estimate reported on arXiv. Area-ratio scaling or transfer of one tropical correction factor to all other vegetation would introduce unsupported density and calibration assumptions. For a strictly forest-only target F , the appropriate partition is instead

$$C_F^{\text{AG}}(t_0) = C_{P \cap F}^{\text{AG}}(t_0) + C_{F \setminus P}^{\text{AG}}(t_0),$$

so the mapped forest contribution must first be distinguished from other included vegetation. Roots, dead organic matter and soil carbon are additional pools, not quantities already represented by the aboveground total.

Table 5 separates predictor acquisition, field calibration and nominal output epoch [1, 16, 24]. Those dates define a multi-date measurement chain; publication alone performs no temporal stock update.

Table 5: Spatial and temporal provenance of the pantropical stock estimate. Publication year, predictor acquisition and nominal output epoch have different meanings. The 2025 global emissions account in Table 4 is a separate annual quantity.

Source / publication	Spatial role	Reference period	Interpretation
Baccini et al. (2012)	Pantropical carbon-map predictor; Australia excluded	Map: 2007–2008; field calibration: 2008–2010	2012 dates the publication, not the mapped carbon.
Simard et al. (2011)	Global canopy-height predictor	GLAS L3C and tree cover: 2005	Predictor coverage and acquisition date do not redefine the carbon output.
Arellano-P. and Rangel-Ch. (2015/2016)	Calibrated pantropical carbon; Australian tropics excluded	Nominal map epoch: 2007–2008	723.97 PgC estimate reported on arXiv; p. 23 specifies the epoch.

Geographic completion and temporal updating are separate operations. On a fixed, consistently tracked pool and mask, moving from t_0 to a later date requires the intervening gains, losses and pool transfers. Extending annual accounts worldwide instead sums the signed regional contributions: $N_W(t) = \sum_d [E_d(t) - U_d(t)]$ over disjoint regions and aligned attribution rules. Additional regions add positive stock and may add both emissions and removals; their net annual contribution can have either sign. Thus a larger worldwide standing stock does not, by itself, establish a larger 2025 net land flux or a proportional change in greenhouse-gas shares.

6 The stock-to-flux transfer identity

6.1 Matched support and fixed release operators

Consider spatial or cohort units $i = 1, \dots, n$ with positive baseline carbon C_i , local correction factors $r_i \geq 0$, and non-negative annual release weights w_i . The units, carbon pools and disturbance histories are held fixed. The weight has dimensions yr^{-1} when a stock is converted to an annual flow. Define

$$\mu_i = \frac{C_i}{\sum_j C_j}, \quad r_C = \sum_i \mu_i r_i, \quad r_F = \frac{\sum_i C_i r_i w_i}{\sum_i C_i w_i}, \quad (10)$$

with $\mathbb{E}_\mu[w] > 0$. Carbon-to- CO_2 conversion cancels in this ratio if applied consistently.

Proposition 6.1 (Stock-to-flux transfer). *Under these conditions,*

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w)}{\mathbb{E}_\mu[w]}. \quad (11)$$

Consequently $r_F = r_C$ if and only if the stock-weighted covariance is zero. Moreover,

$$\min_i r_i \leq r_F \leq \max_i r_i, \quad |r_F - r_C| \leq \sigma_\mu(r) \text{CV}_\mu(w). \quad (12)$$

Proof. Equation (10) gives $r_F = \mathbb{E}_\mu[rw]/\mathbb{E}_\mu[w]$. Substitution of $\mathbb{E}_\mu[rw] = \mathbb{E}_\mu[r]\mathbb{E}_\mu[w] + \text{Cov}_\mu(r, w)$ proves the identity. A positive denominator gives the equivalence. The first bound follows because r_F is a convex combination of r_i with weights proportional to $C_i w_i$. The second follows from Cauchy–Schwarz and $\text{CV}_\mu(w) = \sigma_\mu(w)/\mathbb{E}_\mu[w]$. $\square$

The result is the covariance decomposition of a weighted mean, algebraically identical to the selection term of the Price equation [25]. Its scientific role is to identify the missing empirical quantity: the association between where carbon is misestimated and where that carbon is released. Statistical independence is sufficient but not necessary; zero covariance under the specified stock weights is the exact condition. A constant r_i is another sufficient condition. Neither condition follows from a ratio of published global totals on different masks.

Table 6: Two-unit counterexample. Baseline stocks are (50, 50) and corrections (1, 3) in both rows, so $r_C = 2$. Only the annual release location changes. Units are synthetic.

Annual release weights	$\mathbb{E}_\mu[w]$	$\text{Cov}_\mu(r, w)$	r_C	r_F
(0.02, 0)	0.01	−0.01	2	1
(0, 0.02)	0.01	+0.01	2	3

6.2 Census changes, offsets and state-dependent calibration

For a stable scalar correction, two-census stock change obeys $\Delta C^* = r\Delta C$. A time-varying calibration instead gives the exact decomposition

$$C_1^* - C_0^* = r_0(C_1 - C_0) + C_1(r_1 - r_0). \quad (13)$$

The second term must be interpreted rather than automatically labelled instrument drift. It can reflect a changed calibration procedure, but it can also arise from a fixed size- or age-dependent calibration evaluated at different biological states. A fixed additive offset cancels for an unchanged set of units; recruitment, mortality or support changes can prevent that cancellation. A stable multiplicative error generally changes stock differences. A stand-wide scalar therefore requires justification for heterogeneous trees and changing composition.

This matters for repeated-census forest sinks. Plot-network analyses estimate biomass dynamics and net live-biomass accumulation using repeated tree observations and biomass relationships [26]. A shared, stable multiplicative mass correction changes these inferred gains and losses, not only disturbance emissions. However, a net increase in live aboveground biomass is not gross primary production or automatically net ecosystem–atmosphere uptake. Mortality transfers mass into other pools; decomposition, roots, soils, respiration and exports are needed to close the atmospheric boundary [15]. The paired formulation below applies only after those quantities have been aligned.

6.3 Disturbance cohorts and carbon fate

For area $A_{ia\tau}$ disturbed by process a in year τ , let h index disjoint tree-size, organ and fate compartments, with pre-disturbance carbon density $c_{iah\tau}$ and affected fraction $d_{iah\tau}$. Set $L_{iah\tau} = A_{ia\tau}c_{iah\tau}d_{iah\tau}$. With $k_{iah}^{\text{CO}_2}(t - \tau)$ the annual release rate (yr^{-1}) for CO_2 in year t ,

$$F_{\text{CO}_2}(t) = \frac{44}{12} 10^{-9} \sum_{i,a,h,\tau \leq t} L_{iah\tau} k_{iah}^{\text{CO}_2}(t - \tau). \quad (14)$$

Here L is in MgC and the output is $\text{GtCO}_2 \text{ yr}^{-1}$. For annual bins, $k\Delta t$ is the released fraction with $\Delta t = 1$ yr; cumulative fate fractions must not exceed the available cohort carbon. Other fate fractions represent retained live biomass, dead material, products, charcoal, exports or carbon released in other gases. Their mass bookkeeping must conserve carbon over the specified horizon. Regrowth is a separate uptake term and cannot be represented by making a gross release kernel negative.

The fixed-operator identity is exact when the correction changes carbon amounts but not areas, disturbance fractions or fate kernels. When those inputs also change, r_F must be recomputed from the revised products rather than attributed exclusively to r_C . An index can include cohort and pool, so the identity is compatible with delayed release; it does not remove the need to measure those delays.

6.4 Size-dependent release and the timing of a stock correction

The opposite-order sign used below is Chebyshev's sum inequality for similarly or oppositely ordered sequences, here written under stock weights. Its application concerns physical release from a defined cohort, rather than a rule for booking inventory stock losses. Combustion measurements in a rainforest clearing distinguish markedly different completeness among fuel-size components [27]. This supplies a physical candidate for the covariance in Proposition 6.1, not a universal monotone law: diameter, moisture, decay, burn history and exposure must be controlled. Where structural underestimation is concentrated in large wood and those compartments burn less completely, their carbon correction receives less weight in initial emissions. Extraction of boles into retained products can reinforce the initial delay, provided products and exports are tracked rather than lost from the boundary [15].

Proposition 6.2 (Opposite ordering and cumulative release). *On a fixed affected-carbon cohort with weights $\mu_i = C_i/C_\Sigma$, let $p_{it} \geq 0$ be the fraction of compartment i emitted as CO_2 in annual bin t , and $P_i(H) = \sum_{t \leq H} p_{it}$. If $(r_i - r_j)(p_{i0} - p_{j0}) \leq 0$ for every pair, then, for positive initial release, $r_{F,0} \leq r_C$. For a horizon with positive cumulative release,*

$$r_{F,\leq H} = r_C + \frac{\text{Cov}_\mu(r, P(H))}{\mathbb{E}_\mu[P(H)]}. \quad (15)$$

If $P_i(\infty) = 1$ for every compartment, $r_{F,\leq H} \rightarrow r_C$. The same conclusion holds for any common positive limiting fraction.

Proof. The covariance identity

$$\text{Cov}_\mu(r, p_0) = \frac{1}{2} \sum_{i,j} \mu_i \mu_j (r_i - r_j)(p_{i0} - p_{j0}) \quad (16)$$

has non-positive summands under the ordering assumption. Apply Proposition 6.1 to p_0 and to $P(H)$. A constant positive limiting release fraction has zero covariance. Strict initial inequality holds if some positive-weight pair has a strictly negative product. $\square$

The stock reference here is *carbon affected in the matched cohort*, not all standing pantropical carbon. The premise is a joint ordering to be tested, not a consequence of tree size alone; diameter-dependent drying

or repeated burning can alter it. Complete oxidation as CO_2 within the tracked boundary is stronger than clearing a stand. Persistent charcoal, retained products, export outside that boundary and other gaseous carbon can leave unequal limiting fractions and nonzero cumulative covariance. A delayed correction need not reappear in every subsequent year. Under complete accounting, its cumulative allocation does recover the stock multiplier.

A synthetic two-compartment example uses stocks (50, 50), corrections (1, 3) and first-year fractions (0.8, 0.1). Then $r_C = 2$ but $r_{F,0} = 55/45 = 1.222\dots$. If the remaining fractions (0.2, 0.9) are subsequently oxidized, their combined later multiplier is $145/55 = 2.636\dots$, and the lifetime multiplier is $200/100 = 2$. These synthetic transfers describe a physical release history. Applying them to EDGAR requires identifying whether the target is conversion-year stock-loss booking or atmospheric release: the inclusion of tropical fire CO_2 in Deforestation does not itself make a decay kernel part of the inventory operator. Section 6.5 distinguishes those accounting conventions.

6.5 Annual mixtures, disturbance trends and inventory conventions

A year- t release account combines cohorts of different ages. Assume fixed compartment shares $C_i > 0$ per unit disturbance, fixed corrections r_i , annual fate fractions $p_{i,s} \geq 0$, and a non-negative disturbance history $D(t-s)$. For annual time bins, let

$$W_i(t) = \sum_{s \geq 0} D(t-s)p_{i,s}, \quad r_{F,t} = \frac{\sum_i C_i r_i W_i(t)}{\sum_i C_i W_i(t)} = r_C + \frac{\text{Cov}_\mu(r, W_t)}{\mathbb{E}_\mu[W_t]}. \quad (17)$$

Here $C_i D$ is affected carbon, and division by the one-year bin width gives an annual rate. The common stock weights presume stable compartment composition; changing forest types or disturbance selection require a cohort-specific extension.

Corollary 6.3 (Stationary mixtures and a two-lag trend criterion). *If D is constant over all contributing ages and $\sum_s p_{i,s} = 1$ for every compartment, then $r_{F,t} = r_C$. With two lags, $p_{i,1} = 1 - p_{i,0}$, and $D_0 = D(t)$, $D_1 = D(t-1)$,*

$$r_{F,t} - r_C = \frac{(D_0 - D_1) \text{Cov}_\mu(r, p_0)}{D_1 + (D_0 - D_1) \mathbb{E}_\mu[p_0]}. \quad (18)$$

The denominator is assumed positive. Oppositely ordered corrections and initial releases therefore suppress the annual multiplier during increasing disturbance, enhance it during decreasing disturbance, and cancel under stationarity.

Proof. Stationarity and complete ultimate release make each W_i equal to the same constant. For two lags, $W_i = D_1 + (D_0 - D_1)p_{i,0}$. Substitution into (17) proves the formula and sign. $\square$

For $C = (50, 50)$, $r = (1, 3)$ and $p_0 = (0.8, 0.1)$, stationary clearing gives multiplier 2; doubling the latest cohort gives $255/145 = 51/29$, and halving it gives $172.5/77.5 = 69/31$. The comparison concerns the multiplier relative to r_C , while corrected release remains larger than the baseline. Longer histories use the full convolution. Under conversion-year stock-loss booking, weights instead describe affected area and carbon; EDGAR's Tier 1 description does not specify a size-resolved decay kernel. The physical release operator is needed for atmospheric reconciliation, independently of that inventory convention [21].

Edge-associated biomass contrasts provide an empirical reason to measure the covariance rather than assume it vanishes. Chaplin-Kramer et al. report approximately 25% lower biomass within 500 m of tropical forest edges and estimate nearly 10% overstatement by Tier 1 carbon-stock methods that omit these effects [28]. These findings establish a spatial carbon contrast. If a uniform reference overstates edge stocks and independently observed clearing weights concentrate at those edges, the correction–disturbance covariance is negative. The biomass study alone does not measure that joint clearing distribution or establish a universal global sign.

7 Paired source–sink corrections

7.1 General reconciliation first; common-stock transfer as a special case

Let $F, U > 0$ be a specified source and positive removal magnitude, with corrected quantities supplied by appropriate, potentially different measurement operators. Define $r_F = F^*/F$, $r_U = U^*/U$ and the signed account $N = F - U + Z$.

Proposition 7.1 (Operator-independent net correction). *For any such corrected components,*

$$\boxed{\Delta N = (r_F - 1)F - (r_U - 1)U + \Delta Z.} \quad (19)$$

No common standing-stock basis or size-independent calibration is required.

Proof. Subtract $F - U + Z$ from $F^* - U^* + Z^*$ and substitute the two component ratios. $\square$

A more restrictive representation sets

$$F = \sum_i C_i w_i^+, \quad U = \sum_i C_i w_i^-, \quad w_i^\pm \geq 0, \quad (20)$$

with positive stocks, fixed weights and the same stock correction r_i multiplying both terms *within each unit*. Applying [Proposition 6.1](#) separately gives

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w^+)}{\mathbb{E}_\mu[w^+]}, \quad r_U = r_C + \frac{\text{Cov}_\mu(r, w^-)}{\mathbb{E}_\mu[w^-]}. \quad (21)$$

Consequently, only in that shared-stock fixed-operator case,

$$\Delta N = (r_C - 1)(F - U) + C_\Sigma \text{Cov}_\mu(r, w^+ - w^-) + \Delta Z. \quad (22)$$

The proof is substitution in (19), using $F/\mathbb{E}_\mu[w^+] = U/\mathbb{E}_\mu[w^-] = C_\Sigma$. Spatially heterogeneous r_i remain permissible, but size-dependent change *within* a growing unit cannot generally be replaced by that unit's stock-level multiplier. Positive regrowth from zero stock is outside this representation, and intact-plot increments require endpoint calibration when r changes with state. Thus (19), not (22), governs the general application. Carbon units, atmospheric boundaries and overlap must be reconciled before either is used; a common CO₂ conversion can then be applied.

7.2 Regrowth from zero stock requires an increment basis

Secondary-forest recovery is a trajectory through age, composition and environment, not a fixed fraction of pre-clearing carbon. Neotropical observations document age-dependent and site-dependent biomass recovery after forest removal [29]. Accordingly, let cohort a occupy fixed area A_a , with baseline live-carbon densities c_{a0}, c_{a1} at two ages separated by $\Delta t > 0$. Let $c_{ak}^* = r_{ak} c_{ak}$ denote a specified endpoint recalibration. Areas in hectares and densities in MgC ha⁻¹ give gains in MgC yr⁻¹ when the interval is in years. For a selected non-decreasing baseline trajectory, define $g_a = (c_{a1} - c_{a0})/\Delta t \geq 0$. Its corrected increment satisfies

$$g_a^* = \frac{r_{a1}c_{a1} - r_{a0}c_{a0}}{\Delta t} = r_{a0}g_a + \frac{c_{a1}(r_{a1} - r_{a0})}{\Delta t}. \quad (23)$$

This is an age-cohort application of the endpoint identity, not evidence of temporal calibration drift. In particular, $c_{a0} = 0 < c_{a1}$ gives $g_a^* = r_{a1}g_a$: the cohort is retained and its initially undefined stock ratio need not be estimated.

Proposition 7.2 (Increment-weighted regrowth correction). *For the fixed cohort areas, let $I_+ = \{a : g_a > 0\}$, $I_0 = \{a : g_a = 0\}$, and $K = \sum_a A_a g_a > 0$. Define $q_a = g_a^*/g_a$ on I_+ and $v_a = A_a g_a^*/K$. Then*

$$K^* = \sum_{a \in I_+} A_a q_a g_a + K_0^*, \quad \frac{K^*}{K} = \mathbb{E}_v[q] + \frac{K_0^*}{K}, \quad K_0^* = \sum_{a \in I_0} A_a g_a^*. \quad (24)$$

Proof. Partition the sum $K^* = \sum_a A_a g_a^*$ into I_+ and I_0 , and substitute the definition of q_a on the former. The weights v_a sum to one because zero increments contribute nothing to K . $\square$

The averaging weights are *baseline gains*, not initial stocks. If $K_0^* = 0$, the replacement factor is a gain-weighted mean; a uniform endpoint correction supplies the familiar common-scale special case. A cohort with zero baseline increment but nonzero revised increment contributes additively and cannot be discarded. If $K = 0$, a total multiplier is undefined even when K^* is positive. A newly represented pool with zero baseline carbon at both endpoints also needs an additive model rather than a finite multiplicative correction. Baseline losses and corrected sign reversals require explicitly signed accounting instead of calling every term a removal.

For example, a zero-stock hectare that gains 5 MgC over one year gains 6 after a final-stock correction of 1.2; it remains part of the analysis despite having zero initial carbon. These are illustrative numbers. K is live-carbon accumulation over the selected interval, not gross photosynthesis or a complete atmospheric removal. Regrowth cohorts must be disjoint from any intact-forest accumulation subtotal, and mortality transfers, dead pools, soil exchange and exported carbon must be reconciled before K contributes to an atmospheric account.

7.3 Common-error cancellation and its limits

For a uniform stable multiplier r , fixed Z gives

$$\Delta N = (r - 1)(F - U), \quad \eta = \frac{|F - U|}{F + U}, \quad \frac{|\Delta N|}{|\Delta F| + |\Delta U|} = \eta \quad (r \neq 1). \quad (25)$$

η lies between zero and one. Thus a nearly balanced pair can undergo a large revision of both components with a small *absolute* adjustment of their net difference. The relative change of the nonzero calibrated net difference is still $r - 1$; cancellation must not be described as generally small relative net bias. Exact balance leaves the net unchanged, regardless of the common multiplier.

The converse sensitivity is just as important. Write $r_F = r + \delta/2$ and $r_U = r - \delta/2$. Then

$$N^* = r(F - U) + \frac{\delta}{2}(F + U) + Z^*. \quad (26)$$

A small differential correction is multiplied by the gross sum, not the net difference. Near balance, unequal source and sink corrections can dominate the common-scale contribution and reverse the sign of net exchange. Pairing therefore creates a testable mechanism for hidden gross bias, not an automatic means of making any proposed stock compatible with atmospheric constraints.

A source-reported process pair provides an empirical scale. For 2015–2024, Global Carbon Budget 2025 reports deforestation emissions of approximately 1.9 GtC yr^{-1} and forest (re-)growth removals of 1.3 GtC yr^{-1} [22, Section 3.2.2]. As a selected pair, these yield $F - U = 0.6$, $F + U = 3.2$ and $\eta = 0.1875 \simeq 0.19$; an illustrative $\delta = 0.1$ contributes 0.16 GtC yr^{-1} through (26). It is not the complete net land-use account.

The primary-source decomposition separates permanent deforestation (1.1) and re/afforestation uptake (0.6) from shifting-cultivation emissions and regrowth; subtraction from the rounded totals gives 0.8 and 0.7 GtC yr^{-1} for the latter pair. Table 7 keeps the process definitions explicit. The shifting-cultivation

Table 7: Selected GCB land-use process pairs for 2015–2024 [22, Section 3.2.2]. U is a positive removal magnitude; rates are in GtC yr^{-1} . Shifting-cultivation values are derived from rounded aggregates. The final column is the synthetic sensitivity $0.1(F + U)/2$, not an observed calibration correction.

Pair	F	U	$F - U$	η	Shift at $\delta = 0.1$
Selected total	1.9	1.3	0.6	0.19	0.160
Permanent change	1.1	0.6	0.5	0.29	0.085
Shifting cultivation	0.8	0.7	0.1	0.07	0.075

entries are differences of rounded source estimates; their uncertainties cannot be derived by subtracting marginal model ranges.

The two-lag and finite-increment results explain why shared land does not imply identical annual correction factors. A shifting-cultivation cycle can share a carbon calibration across clearing and regrowth, but vegetation age, organ allocation and lagged cohorts can still produce $r_F \neq r_U$. Permanent loss and re/afforestation need not involve the same regions or populations. GCB identifies China, the USA and the EU27 as leading re/afforestation removal contributors, with 0.3 GtC yr^{-1} combined—about half of the rounded permanent-removal total [22, Executive summary and Section 3.2.2]. Country boundaries do not identify tropical or extratropical pixels; a tropical-only recalibration cannot be assigned uniformly to that global sink.

Let p_F and p_U be the fractions of the selected source and removal affected by a local common multiplier r , leaving their complements unchanged. Then the specified scenario gives

$$\Delta N = (r - 1)(p_F F - p_U U),$$

where the fractions are *flux-weighted exposure*, not simply shared land area. Unequal exposure produces the differential case even for one locally shared multiplier. For example, $r = 1.2$ on the permanent pair with $p_F = 1$, $p_U = 0$ changes the net by 0.22, compared with 0.10 when both are fully exposed. These are testing scenarios, not inferred geographic fractions. Process- and region-resolved carbon calibration determines which scenario is admissible; neither cancellation nor source-only amplification is imposed on the actual inventory.

The EDGAR entries in Table 4 are signed reporting categories with a different boundary; their opposite signs do not make them the same gross process pair. The GCB decomposition is used here specifically to separate the above mechanisms.

7.4 Size-dependent calibration and a directional census test

Size- and architecture-dependent discrepancies motivate a differential-correction test across all represented stem classes [1, 19, 30]. Large-tree studies provide one calibration domain, while the intermediate-stem contributions in Section 4.4 require their own event weights. Neither individual tree size nor a selected validation sample establishes the dominant contribution to population-level release, or that mortality receives a larger correction than growth. To see the distinction, let x be a tree’s baseline carbon and let a stable, continuously differentiable, zero-intercept calibration be $m(x) = r(x)x$. A surviving tree growing from x_0 to $x_1 > x_0$ has gain multiplier

$$q_G(x_0, x_1) = \frac{m(x_1) - m(x_0)}{x_1 - x_0}, \quad m'(x) = r(x) + xr'(x). \quad (27)$$

The secant equals the average of m' over that interval. Mortality of a tree with assigned pre-death stock x_d , in contrast, contributes $m(x_d)/x_d = r(x_d)$. An increasing stock multiplier is not the same function as the growth multiplier. Species, architecture and changing organ allocation can require a multivariate calibration; the scalar construction isolates the logical distinction without imposing a universal fitted curve.

Let $G, L > 0$ denote positive gains and losses in a matched live-carbon census account, with net accumulation $S = G - L$. Sum calibrated survivor increments and consistently accounted recruitment into G^* ; use matched pre-death stocks and other specified live-pool losses for L^* . The resulting ratios $r_G = G^*/G$ and $r_L = L^*/L$ give

$$S^* = r_G G - r_L L = r_G S - (r_L - r_G)L, \quad S^* - S = (r_G - 1)G - (r_L - 1)L. \quad (28)$$

Recruitment across an inventory threshold is an entry into the observed pool, not necessarily carbon assimilated wholly during that interval. Diameter thresholds, survivor shrinkage, missing stems and death-time conventions must therefore be fixed or reconciled. Mortality here is loss from the live pool; atmospheric release follows its carbon-fate operator, not immediate emission by definition.

Directional hypothesis H_Δ . In a specified, independently calibrated census population, the event-weighted loss correction exceeds the gain correction by a preregistered meaningful margin, $r_L - r_G > \epsilon_\Delta \geq 0$. This predicts $S^* < r_G S$, lower net live-carbon accumulation than under common gain scaling. For S^* to be smaller than the *uncorrected* sink, the exact condition is $(r_L - 1)L > (r_G - 1)G$. When $S > 0$ and $r_G \geq 1$, this condition implies the zero-margin directional condition $r_L > r_G$; it need not imply an independently chosen positive margin ϵ_Δ . Without those conditions, calling it a universally stronger inequality is unwarranted. Neither prediction follows solely from larger average size among dead trees. The hypothesis concerns measured event-weighted corrections. A prespecified interval for their difference wholly above the margin supports the directional claim; an interval wholly at or below the margin opposes it, while overlap leaves the test unresolved at that precision.

Table 8: Synthetic differential-census cases with $G = 120$, $L = 100$ and $S = 20$ in arbitrary carbon units over a common interval. The second row uses one increasing size-calibration function, not fitted forest observations.

Case	r_G	r_L	G^*	L^*	S^*
Loss-weighted hypothesis	1.20	1.50	144.0	150.0	-6.0
Increasing-size counterexample	1.85	1.50	222.0	150.0	72.0

The counterexample in Table 8 uses $m(x) = x(1 + 0.005x)$ in the specified synthetic units: 12 survivors grow from 80 to 90, while one larger tree of stock 100 dies. Then $r_G = 1.85 > r_L = 1.50$ and $S^* = 72$, despite the strictly increasing $r(x)$ and the larger dead tree. Underestimation of large stock levels therefore does not determine the corrected sink's direction. This counterexample does not deny differential amplification; it identifies the finite-increment evidence needed to turn that mechanism into a defensible prediction.

7.5 Operational census screens

For a common calibration applied to a matched census, let $\bar{x}_L$ be the mortality-stock-weighted mean tree carbon and $\bar{x}_G$ the gain-weighted mean of survivor interval midpoints. The quadratic family $m(x) = \alpha x + kx^2$, with $k > 0$, corrects losses more than gains exactly when $\bar{x}_L > 2\bar{x}_G$. A specified positive effect margin additionally requires the fitted magnitude of k .

For $m(x) = ax^b$, $b > 1$, the exact finite-interval screen instead uses the ratio of loss- and interval-gain-weighted $(b - 1)$ -power means, which must exceed $b^{1/(b-1)}$. An arithmetic ratio below two excludes the quadratic direction but does not exclude every power exponent. Near $b = 1$, the log-moment difference determines the first-order direction and the differential tends to zero. Supplementary Note S2 gives the definitions, proofs, threshold behaviour and counterexample. These screens use baseline event distributions to identify admissible directions; independent calibration and uncertainty remain necessary to estimate their size.

7.6 Gross-source and net-land shares are different quantities

For a simplified account with other positive sources O and $Z = 0$, define

$$s_F^{\text{gross}} = \frac{F}{O + F}, \quad s_N^{\text{net}} = \frac{F - U}{O + F - U}. \quad (29)$$

The first is the forest fraction of the selected positive source sum. The second is a signed net-land contribution relative to a net total; it can be negative and becomes unstable as that total approaches zero. $F/(O + F - U)$ is a third ratio: a gross component divided by a net total. It is not a conventional composition of non-negative shares and can exceed one. When Z contains additional sources and removals, those components must be explicitly partitioned before forming a gross-source sum.

Table 9: Synthetic paired-correction example: $F = 6$, $U = 5.8$, $O = 50$, $Z = 0$. Flow units are arbitrary common annual mass units, not 2025 observations. The last two columns follow (29); percentages have different denominators.

Case	r_F	r_U	F^*	U^*	$F^* - U^*$	Gross share (%)	Net share (%)
Reference	1.0	1.0	6.00	5.80	0.20	10.71	0.40
Source only	2.5	1.0	15.00	5.80	9.20	23.08	15.54
Paired uniform	2.5	2.5	15.00	14.50	0.50	23.08	0.99
Paired differential	2.5	2.6	15.00	15.08	-0.08	23.08	-0.16

In the uniform example, corrections of +9 and +8.7 in gross emission and removal magnitudes change the net by only +0.3. The cancellation ratio is about 0.01695. With a slightly larger sink multiplier, the revised net becomes negative. These are algebraic demonstrations of identifiability and denominator dependence, not inferred global forest fluxes. They motivate independent constraints on source and removal weights, rather than a preferred multiplier.

7.7 Independent measurements remain decisive

A common allometric calibration can affect census-derived quantities, but it need not affect an independently measured atmospheric concentration increment, fossil-fuel statistic or tracer constraint in the same way. The theoretical cancellation concerns shared calibration of specified components; regrowth and live-census accumulation enter only after their increment operators and pool boundaries have been aligned. It cannot be imposed on independently constrained terms merely to force closure.

Likewise, living-biomass uptake, managed-land net removals and bookkeeping net land-use emissions are not interchangeable sink variables. A correction must follow the underlying mass estimator into its valid pool and process, account for time lags, and identify overlap with other entries. Only then can the proposed source–sink revision be tested against a carbon-mass budget. The fact that a budget can contain compensating errors supplies no evidence for a particular compensation [22]. Without a supported offset, an upward revision of actual net forest release increases the corresponding reconstructed atmospheric accumulation, as made explicit in Section 8.1.

8 Atmospheric consequences and source attribution

8.1 Omitted net sources and the observed carbon balance

For atmospheric carbon mass M_{atm} , a harmonized account obeys

$$\frac{dM_{\text{atm}}}{dt} = E_{\text{DD}} + E_{\text{other}} - U_{\text{land}} - U_{\text{ocean}}.$$

Here E_{DD} is the defined deforestation/degradation release and U denotes positive removal magnitudes. For corrected-minus-reference differences with the same initial mass,

$$\delta M_{\text{atm}}(t_1) = \int_{t_0}^{t_1} [\delta E_{DD} + \delta E_{\text{other}} - \delta U_{\text{land}} - \delta U_{\text{ocean}}] dt.$$

A missing positive net source places actual accumulation above the account's reconstruction unless other terms offset it. Delayed decomposition assigns release to later years; reduced uptake enters once through the relevant removal term. Direct atmospheric observations may already contain carbon absent from a bottom-up inventory, in which case the correction changes its source attribution rather than the observed concentration [22, 31].

8.2 Accounting alternatives and their empirical standing

For the signed account $T = F + O - U$, the source, other-source and removal corrections satisfy

$$\Delta T = \Delta F + \Delta O - \Delta U. \quad (30)$$

The three outcomes in Figure 2 retain the substantive inference of forest-carbon recalibration: an independently supported upward correction to annual deforestation and degradation can increase the estimated total, increase the contribution assigned to forest loss within a retained total, or change several sources and removals together. They are conditional cases of one balance, rather than competing physical theories or three measured present-day allocations.

I. Larger total with other components retained. For $\Delta F > 0$ and $\Delta O = \Delta U = 0$, the total rises by $\Delta T = \Delta F$. On a physical carbon-mass boundary this yields greater reconstructed atmospheric accumulation over the corresponding release history. Delay redistributes release among years; it does not remove the later contribution. This case tests an omitted source, without assuming a compensating sink.

II. Greater forest attribution within a retained total. If the total and removals are independently retained, $\Delta T = \Delta U = 0$ implies $\Delta O = -\Delta F$. The forest contribution increases and the aggregate assigned to other sources decreases. Evidence must identify the affected other components; the identity does not assign the adjustment to fossil energy by default. A leading forest-loss contribution is a conditional outcome, requiring $F^* > \max_j O_j^*$ for positive source categories on a common boundary. A majority requires the stronger condition $F^* > \sum_j O_j^*$. An increase in share does not itself establish either ranking.

III. Joint source–removal revision. Carbon calibration can affect both release and accumulation, and multiple terms may require reassessment. Their supported corrections enter the same signed identity. The net adjustment can be positive, negative or small; shared calibration does not establish equal multipliers or an automatic cancellation. This case accommodates the measurement dependence developed in Section 7 without prescribing its empirical result.

Largest-sector inference with the other components retained. A change in total and a change in sector ranking can occur together under case I. Let the consistently defined land category be $L = F + Z_L$, with its other net contribution Z_L retained, and let O_j be unchanged peer sectors. Then $L^* = r_F F + Z_L$ and, for a common positive revised total,

$$L^* > \max_j O_j \quad \Leftrightarrow \quad \frac{L^*}{T^*} > \max_j \frac{O_j}{T^*}.$$

Thus the corrected land category *is the largest sector* whenever its corrected absolute contribution exceeds every retained peer contribution. Renormalizing the percentages cannot undo that ranking. Keeping the other components means keeping their absolute amounts and accounting boundaries, not keeping the

total or their old percentages simultaneously. The sufficient and necessary threshold for positive F is $r_F > (\max_j O_j - Z_L)/F$.

Figure 2 gives a conditional numerical instance on Ecofys's 2010 grouping, alongside the questioned total and double-headed arrow. The source's land-use group combines the 10.3% deforestation/afforestation and land-use CO₂ component with 4.4% agricultural soils; its industry subflows sum to 29.1%, the largest comparison sector [5]. Imposing the historical ratios 723.97/285.1 and 723.97/228.7 on the 10.3% component, while retaining all other absolute contributions, gives land-use-group shares of 26.37% and 30.26% of their respective revised totals. The retained industry shares become 25.12% and 23.79%. The land-use group therefore ranks first at both endpoints in this reconstruction, without reducing industry's absolute emissions. This historical grouping includes agricultural soils and differs from modern EDGAR LULUCF; the component alone exceeds industry only at the upper endpoint. The empirical stock-to-annual transfer and source compatibility remain to be established, as specified in Supplementary Note S3. Neither a present-day LULUCF ranking nor dominance over the combined energy supersector follows from this historical illustration.

The reference labels on the non-land branches of Figure 2 identify baseline contributions, not exact or unchangeable values. Its highlighted outcomes concern the propagation of measurement corrections; graphical prominence is not a quantitative ranking. Gross forest release, net land-use exchange and multi-gas sector accounts are different comparisons and require their own source definitions.

For a specifically defined 2024 two-source CO₂ account, GCB gives fossil emissions 10.3 and net land-use emissions 1.3 GtC yr⁻¹ [22]. Reassigning x from fossil to net land use would reverse their ranking only if

$$1.3 + x > 10.3 - x, \quad x > 4.5 \text{ GtC yr}^{-1}.$$

That exceeds nine times the quoted 0.5 GtC yr⁻¹ fossil standard-uncertainty scale and implies a fossil reduction of approximately 44%. This particular fossil-to-net-land ranking reversal is unsupported by the cited evidence and cannot be inferred from a stock ratio. The calculation does not reject every fixed-total correction: a smaller reattribution, a correction within specified land components, or a comparison of gross forest release with disaggregated sectors is a different claim requiring its own observations. In particular, gross release accompanied by removals cannot be substituted for the net land-use term in this two-source benchmark. A quoted standard uncertainty is not a hard physical bound or a Gaussian tail probability.

8.3 Persistent corrections and joint constraints

Define the carbon-budget imbalance $B = E_{\text{fossil}} + F - U - S_{\text{ocean}} - G_{\text{atm}}$. Its change is

$$\Delta B = \Delta E_{\text{fossil}} + \Delta F - \Delta U - \Delta S_{\text{ocean}} - \Delta G_{\text{atm}}.$$

The baseline residual, period and joint uncertainty determine admissibility. Selected source constraints are in Table 10. GCB describes the 1959–2024 budget as approximately balanced overall, while reporting a cumulative residual of about +35 GtC, roughly 7% of total emissions; the 2015–2024 mean is below 0.1 GtC yr⁻¹. Its –1.7 GtC residual in 2024 instead suggests overestimated total uptake or atmospheric growth that year [22, Sections 3.5 and 3.9]. These temporal constraints are distinct. An uncompensated +0.5 GtC yr⁻¹ source over 66 years adds 33 GtC to the positive cumulative residual, worsening it to about +68 GtC. Offsets of –33 and –68 GtC would, respectively, preserve the baseline residual and close that constructed account to zero; neither is established by this sensitivity calculation.

These constraints limit particular magnitudes and histories, rather than eliminate the three outcomes in Figure 2. The illustrated large persistent source-only addition worsens the retained positive cumulative residual; the illustrated several-GtC fossil-to-net-land reassignment lacks independent support. Other

Table 10: Selected Global Carbon Budget 2025 constraints used to assess source-account alternatives [22]. Annual rates are in GtC yr^{-1} . Quoted uncertainties are source estimates, not admissibility ceilings. Annual and decadal estimates are kept distinct.

Quantity	Period	Reported estimate / implication
Fossil emissions	2024	10.3 ± 0.5 ; a large downward correction requires independent evidence.
Atmospheric growth, surface record	2024	7.9 ± 0.2 ; use the detailed atmospheric section.
Atmospheric growth, decadal mean	2015–2024	5.6 ± 0.02 ; not the uncertainty of a single year.
Net land-use emissions	2015–2024	1.4 ± 0.7 ; distinct from gross deforestation.
Environmentally driven land sink	2015–2024	2.4 ± 0.8 ; different attribution from land-use regrowth.

periods, spatial domains and component corrections must be evaluated against their corresponding evidence. Joint land-process reconciliation is appropriate when biological calibration affects several components, but is not an assumed resolution of an upward forest-source correction. It still requires independent source, removal and regional constraints; it is not guaranteed compensation. Common scaling of a nearly balanced pair gives a small absolute net change, whereas the differential term $\delta(F + U)/2$ can produce a large net change or sign reversal. The constraint is the signed multiyear sum and its uncertainty, not a rule that only small-net paired revisions are physically possible.

9 A calibration-first programme for forest-carbon accounting

9.1 Independent material comparisons

Compare allometric and reconstructed carbon with a common, independently checked reference on matched trees, plots, pools and dates. Retain organ geometry, basic density, tissue fractions, nonmerchantable material and a distinct dead-pool account. For reference C_{ref} , report residuals $e_A = \hat{C}_A - C_{\text{ref}}$ and $e_R = \hat{C}_R - C_{\text{ref}}$ with their joint uncertainty and dependence on architecture, composition and disturbance. A negative allometric residual associated with missing architectural predictors supports the underestimation mechanism; each pathway’s residual remains subject to the same decision rule.

The class-5 application has neither a recovered independent geometric residual nor a validated joint physical envelope. The Córdoba aggregates lack sound organ volumes and matched map values. Supplementary Note S1 identifies the structural combinations needed to attain the mean and the conditions under which an independently measured upper envelope would reject it. A prespecified equivalence margin permits agreement, positive or negative discrepancy, or an unresolved comparison. Representative area weighting, held-out sites and material-property uncertainty distinguish empirical evaluation from numerical closure of the reconstruction identity.

9.2 Calibration, spatial transfer and disturbance

Build vegetation-specific reference distributions and withhold complete sites before label construction, fitting or model selection. Record the joint predictor support and original observations, including acquisition dates and procedural reconstruction steps. Diagnose saturation using independently measured material; a response mean beyond the historical label maximum alone does not characterize every pixel’s predictor support. Refinement of material labels and their transport can change the estimated stock without changing the general label-inheritance identities.

Sample carbon where deforestation and degradation occur, retaining intermediate stems, large trees, partial crowns, collateral mortality and repeated damage. Apply affected fractions once to the remaining

carbon, then track release, products, dead material, exports and regeneration separately. The relevant association is between carbon-calibration error and the actual source/removal weights. Apply census calibration to both survivor endpoints and the specified mortality stocks, with recruitment and measurement thresholds explicit. The screens in Section 7.5 identify family-specific directions; independent calibration estimates their magnitudes.

9.3 Independent reconciliation

Reconstruct physical source/removal histories before assigning them to inventory categories. Aligning managed-land and environmental-change conventions can materially alter an aggregate comparison [23]. On a common carbon-mass boundary,

$$G_{\text{atm}} = E_{\text{fossil}} + F - U - S_{\text{ocean}} + R, \quad (31)$$

fixing all other terms identifies $F - U$, while $(F, U) \mapsto (F + a, U + a)$ preserves that balance. Regional, temporal and tracer observations can constrain the remaining direction [32]. Validation against independently measured components is therefore distinct from forcing an aggregate to close.

10 Discussion and conclusions

The framework connects three operations: biological calibration, spatial prediction and annual transfer. A label-trained LiDAR predictor inherits the conditional reference error; changing architectural composition can change a transferred calibration; and release-weighted corrections depend on disturbance and carbon fate. The paired and census operators make source/removal and gain/loss dependence explicit. Their assembly gives a practical validation order: establish material carbon, test spatial transport, and constrain annual exchange with independent observations.

The physical comparison also gives a definite result. Under the illustrative material properties, low-cylinder configurations yield less carbon than the initial class-5 target. Attaining that target requires larger joint basal-area, height and crown-volume products, different material properties, or a revised map. However, a mean crown-mass fraction is neither a whole-tree volume ratio nor a class-wide upper bound. The harvest and TLS literature supports architectural variability and correction of individual-tree estimators; it does not establish the representative distribution needed to validate the initial class mean or the joint bound needed to reject it physically. That magnitude remains unverified, rather than established by the general allometric-bias mechanism.

The empirical claims have distinct scales. Organ measurements and tissue analyses support the original reconstruction chain at their documented domains. The source-table correction changes printed class assignments, leaving carbon stocks unchanged. The support-cap calculation quantifies a numerical operation, not the error of the entire map. All stem sizes enter source populations through abundance, affected fractions and subsequent fates; a sparse large-tree sample cannot establish that distribution.

Figure 2 expresses three conditional consequences of improved forest-carbon measurement: a larger source total when other components are retained, a larger forest contribution when the total and removals are retained, or a joint revision of sources and removals. These alternatives remain analytically valid and empirically testable. A leading forest-loss contribution requires the specified comparison among annual sources; it is neither proved nor universally excluded by the standing-stock estimate. Independent fossil and atmospheric constraints restrict the magnitude and history of any reattribution. The specific two-source net-ranking reversal calculated here lacks the required fossil evidence, while persistent unoffset source additions must fit multiyear balances. Paired correction is one admissible structure, not compulsory cancellation: differential errors act on the source–removal sum. Continued calibration can therefore refine both the estimated carbon and its annual consequences without fixing their outcome in advance.

The ranking implication is affirmative and conditional: when a supported correction makes the defined land-use-and-forestry contribution exceed the largest retained peer sector, it is the largest compartment of that account, even after the increased total is used to renormalize every share. Figure 2 illustrates this property under the explicitly stated 2010 Ecofys grouping; the contemporary net-LULUCF account and the combined energy supersector are different comparisons.

Data and code availability

The V7.5 companion author archive contains the article, technical supplement, figure plates, transcribed inputs and executable analyses, with an edition-specific SHA-256 manifest. Numerical verification regenerates 76 outputs. Public editions are maintained separately in the [companion research repository](#); the existing V7.0 deposit has its own publication manifest, while this revision is supplied as a companion archive. Editable document sources are excluded from the public repository distribution. Original-record limitations and capture-route fields are documented in `SOURCE_AVAILABILITY.md` and `data/capture_route_schema.json`. No archival DOI has been assigned to this revision.

Author contribution

Henry Arellano-Peña is the sole author and is responsible for the analysis, interpretation and presentation.

Competing interests

Henry Arellano-Peña owns the methodologies discussed and is founder and director of NEBIOT S.A.S., the sole company affiliated with this work. NEB001 provides the vegetation-mapping framework and NEB002 the architecture, material-analysis and calibration chain underlying the initial arXiv reassessment [1, 2]. These relationships are disclosed; the author declares no other competing interests.

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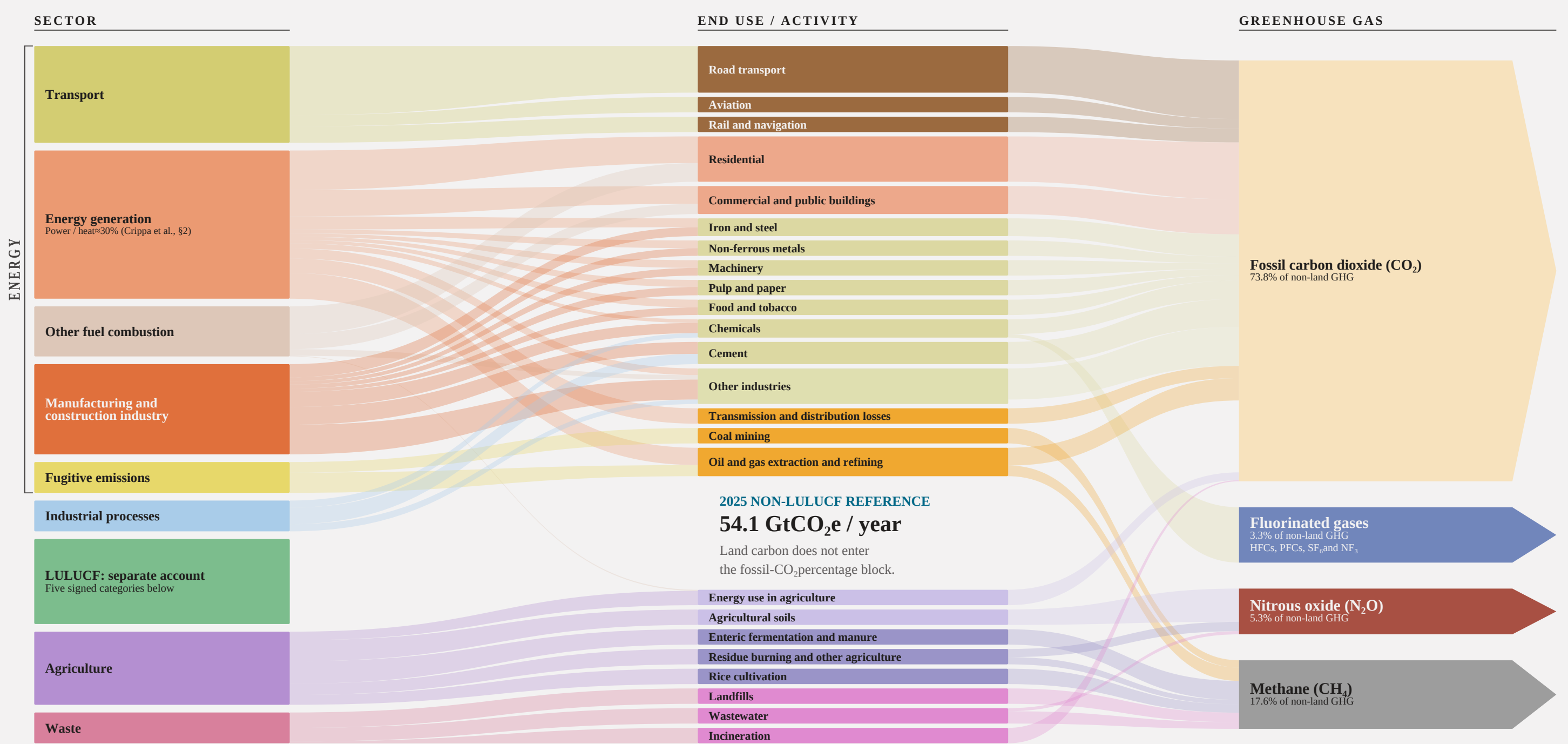
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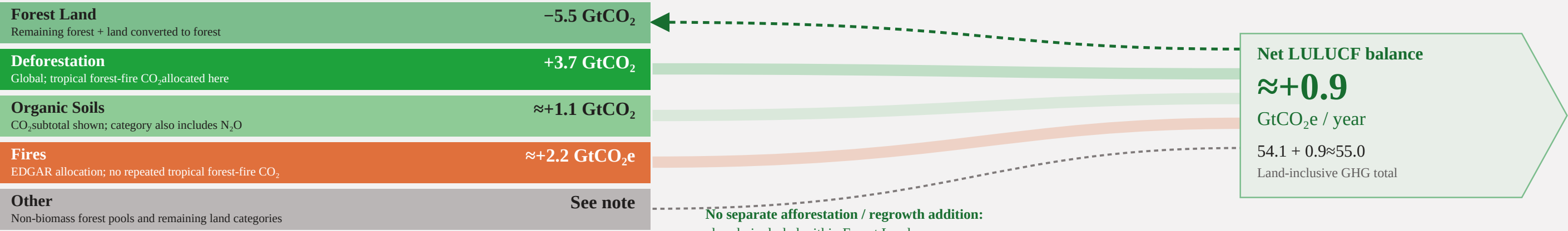
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THE SIGNED LAND ACCOUNT

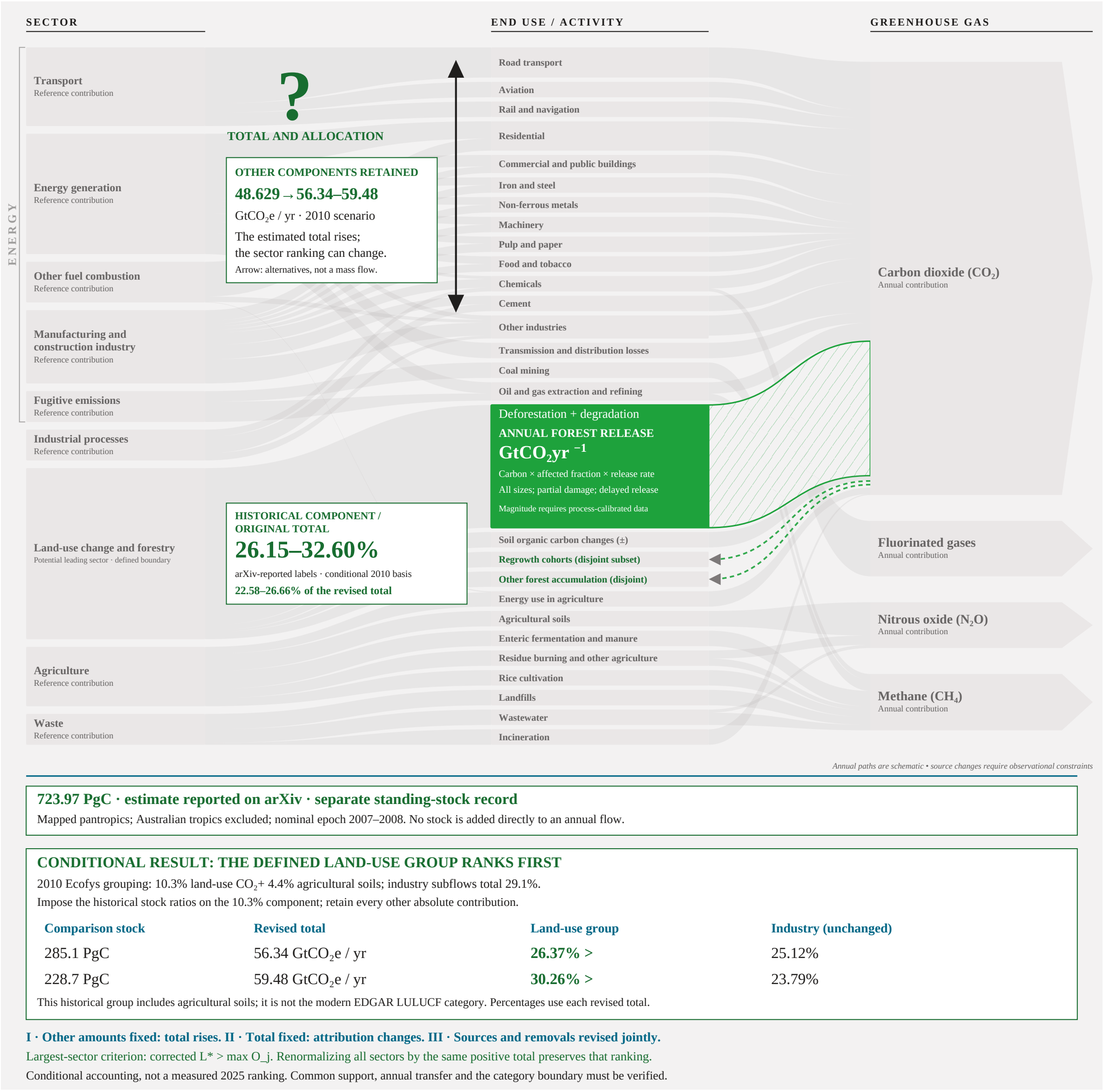
EDGAR 2026 · Section 3, Figure 6 and Annex 2 · 2025 estimates



Other / rounding remainder:≈-0.6 GtCO₂e is a closure diagnostic, not an independently extracted category total.

Disjoint land categories, separate gas denominator. Forest Land already includes land converted to forest; no extra afforestation or regrowth subtotal is added. Deforestation includes tropical forest-fire CO₂, which is excluded from Fires; tropical fire non-CO₂ gases remain in Fires. Organic Soils’ quoted 1.1 is CO₂ only. The diagnostic remainder 0.9 − [3.7 − 5.5 + 2.2 + 1.1] ≈ −0.6 also absorbs omitted subcomponents and rounding; it is not a separately extracted Other estimate. Source: Crippa et al. (2026), booklet Sections 2–3 and Annex 2, doi:10.2760/7717504 (CC BY 4.0).

Figure 1: The reported 2025 account and its boundaries. The sector–activity–gas organization is adapted from WRI (2005) and Ecofys (2013), with author-created vector artwork and a separate five-category signed land account. Intermediate activity nodes describe pathways, rather than EDGAR’s official sector list. Gas shares refer only to EDGAR’s 54.1 GtCO₂e non-LULUCF inventory using AR5 GWP₁₀₀; 73.8% is fossil CO₂. Land emissions therefore do not enter that gas block. The reverse dashed link denotes Forest Land’s net removal. Forest/deforestation estimates use a global Tier 1 approach with default parameters and estimated activity data beyond the latest input maps, not independently remeasured 2025 carbon densities. Paths and all graphical areas are schematic, not a balanced quantitative Sankey. Electricity-attribution paths must not be counted again as direct emissions.



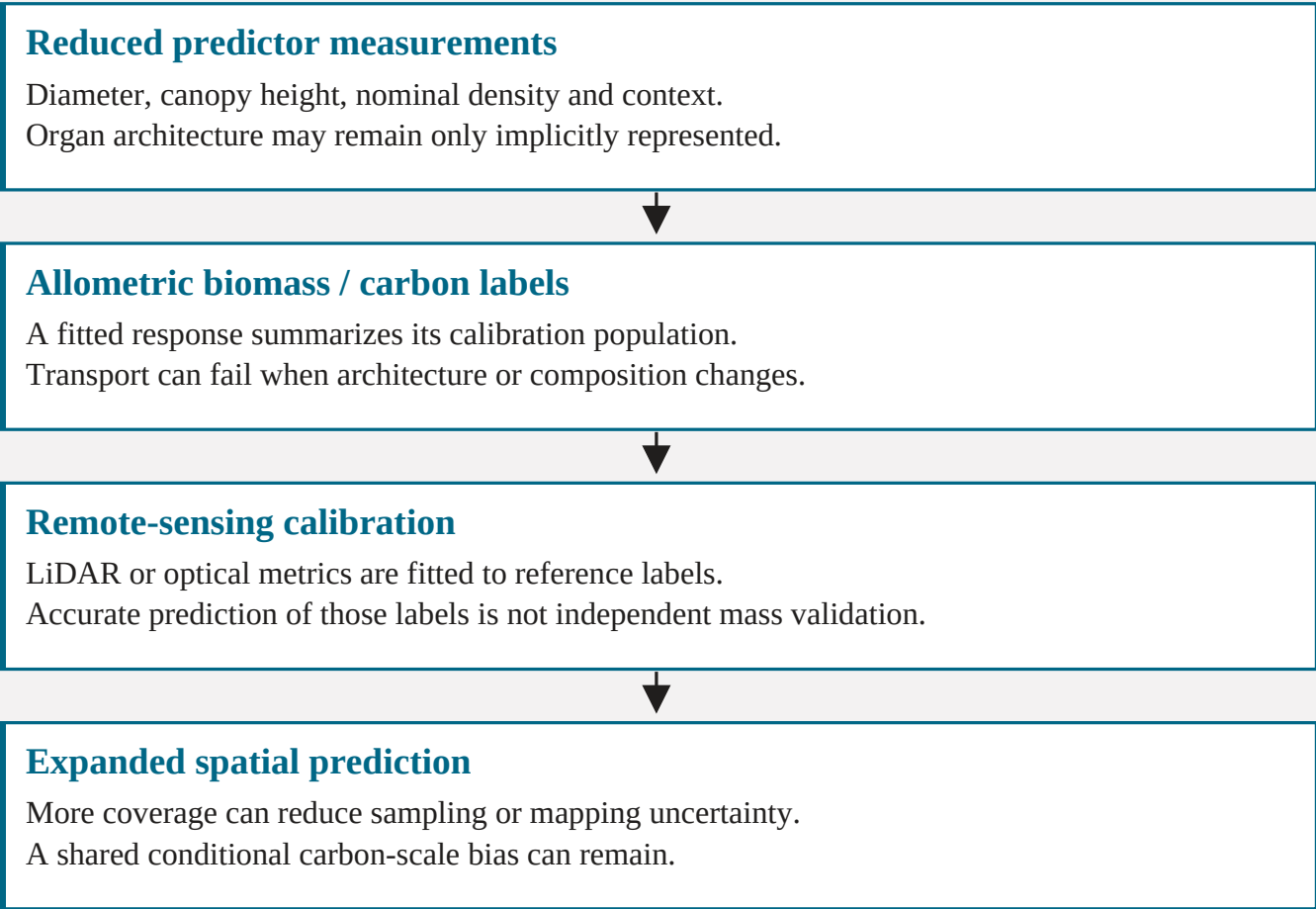
All stem sizes and explicit carbon fate. Per-tree carbon, abundance, affected fractions and release timing determine annual deforestation and degradation. An omitted positive net source leaves actual accumulation above the account’s reconstruction unless other changes offset it. Observed air can already contain emissions missed by a bottom-up inventory. Carbon released after clearing remains in its later-year balance; no unmeasured compensating sink is imposed.

Figure 2: The numerical second interpretation and its conditional largest-sector inference. The question mark and double-headed arrow concern the estimated total and sector allocation, not an established error in a named energy source. The historical annotation restores the arXiv-reported 26.15–32.60% range; nearest rounding from displayed inputs gives 26.16–32.61%. Under the imposed historical annual multiplier and retained other absolute contributions, totals become 56.34–59.48 GtCO₂e yr⁻¹. In Ecofys’s 2010 grouping, adding the retained 4.4% agricultural-soils component makes its land-use group exceed the 29.1% industry subflow sum at both endpoints, before and after renormalization. That grouping differs from modern EDGAR LULUCF and from the conceptual sector layout above; the numerical card is a separately defined historical example, not a measured 2025 allocation. The component alone ranks above industry only at the upper endpoint. Supplementary Note S3 states the transfer, source-provenance and denominator assumptions. The 723.97 PgC stock remains separate from the annual network. All widths and arrows are schematic; the energy supersector is not a single peer category in the ranking. Layout adapted from WRI (2005) and Ecofys (2013); source interpretation: Arellano-P. and Rangel-Ch. (2015/2016).

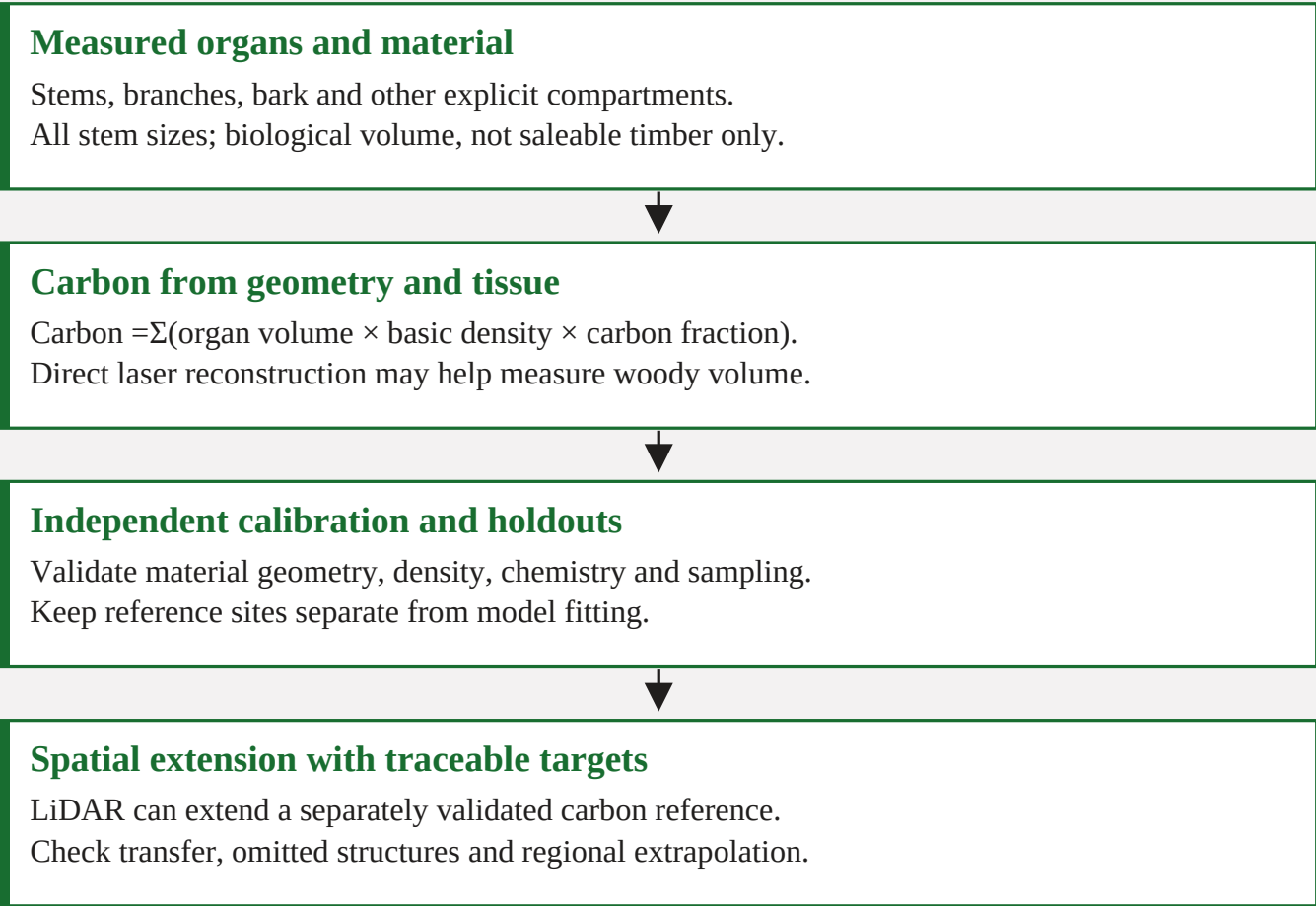
CARBON ACCURACY STARTS WITH THE REFERENCE

Spatial coverage measures where; a validated material conversion determines how much.

A • PROXY-LABEL PATHWAY



B • ARCHITECTURE-INFORMED PATHWAY



INHERITANCE IDENTITY • IDEAL CONDITIONAL PREDICTION

$E[\text{label} \mid \text{LiDAR}] - E[\text{true carbon} \mid \text{LiDAR}] = E[\text{reference-label bias} \mid \text{LiDAR}]$

Conditional mean-zero residual noise is assumed. This is a mechanism, not an empirical global correction factor.

Initial estimate reported on arXiv

723.97 PgC · architecture-informed calibration
Nominal map epoch 2007–2008; Australian tropics excluded.
221 reconstructed trees; 2,232 tissue samples reported.
Successor calibration in development; no new total reported.

From carbon measurement to annual accounts

All sizes × abundance × affected fractions × release histories.
Deforestation, degradation, regrowth and other removals.
Uncounted net release raises reconstructed accumulation.
A retained total constrains reattribution among sources.

NOT EVERY LiDAR PRODUCT USES THE SAME MEASUREMENT CHAIN

Disney et al. (2018); Calders et al. (2022); Wilkes et al. (2023): independent TLS reconstruction pathways.

Measure architecture and organ allocation to identify the source of allometric calibration error.

Refining calibration can improve the stock estimate without changing the distinction between reference accuracy and coverage.

Sources: Arellano-P. & Rangel-Ch. (2016); Goodman et al. (2014); Calders et al. (2015); GEDI L4A V2 documentation.
Operational diagram: no graphical width represents a measured carbon share. Henry Arellano-Peña.

The calibration distinction. A remote-sensing product fitted to allometric reference labels can inherit their conditional bias. Direct reconstruction of biological material with measured tissue properties provides a different reference pathway, which can itself use terrestrial laser measurements. Independent reference tests are required for both approaches; greater coverage alone does not determine absolute carbon accuracy.

Figure 3: Two measurement-to-map pathways and their annual-accounting consequences. The left pathway extends a calibrated proxy; the right begins with organ geometry and matched material properties. Architecture and tissue variation identify mechanisms by which an allometric predictor can depart from actual biological carbon. The conditional-expectation identity assumes a source distribution and mean-zero residual noise conditional on the sensor variables; a real learner also has fitting and transfer errors. The initial reconstruction and physical tests are reported in the arXiv study of Arellano-P. and Rangel-Ch. (2015/2016). Crown-related biomass discrepancies are independently illustrated by Goodman et al. (2014). Disney et al. (2018), Calders et al. (2015, 2022) and Wilkes et al. (2023) document terrestrial laser material reconstruction and segmentation, while the GEDI L4A Version 2 guide documents a model linking waveform height metrics and biomass reference data. These examples retain their study domains and establish no universal multiplier. Total material volume includes nonmerchantable woody organs; belowground and dead pools remain separately defined. The 723.97 PgC value is the initial estimate reported on arXiv. Its annual implications require affected populations, release histories and removals. Neither larger spatial coverage nor numerical reproduction of calibration labels is equivalent to independent carbon-mass validation. Original diagram: Henry Arellano-Peña.

Technical supplement

Allometric calibration bias in tropical forest carbon

Derivations, material-accounting sensitivities and source-table provenance

Henry Arellano-Peña

NUEVO ESTANDAR BIOTROPICAL, NEBIOT S.A.S., Bogotá, Colombia

18 September 2026

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This supplement supplies the detailed calculations supporting the article’s measurement-to-accounting argument. Conditional material requirements, source-table corrections and synthetic tests are distinguished from field observations. The article’s initial-estimate status box defines the source application; the structural comparisons below are conditional tests rather than a new field census. Unprefixed references to article equations use the accompanying compiled article; supplement equations and tables carry an S prefix.

S1 Material-accounting definitions and conditional class-density comparisons

The protocol has four linked operations: fix the spatial and carbon-pool target; independently measure geometry and material properties; reconstruct carbon using disjoint compartments and representative plot weights; and compare the resulting class mean with the mapped target under joint uncertainty. Its empirical test is the comparison between independently obtained estimates. Re-expressing one set of measurements in equivalent algebraic forms provides internal consistency, rather than independent validation. The carbon assigned to a mapped class can be translated into material-volume requirements. Class 5 in the source table contains 285.03 PgC over 531,449,420.52 ha, implying $536.3257 \text{ MgC ha}^{-1}$ [1, Table 3]. Its stock is numerically 99.975% of Feldpausch et al.’s entire 285.1 PgC total, on an area 31.871% as large as that comparison’s 1667.5 million hectares [2]. The area is 5.31 million km^2 , comparable in scale to the approximately 5.3 million km^2 of Amazonian tropical forest described by Fauset et al. [3]; this is a historical scale comparison, not a claim of matching masks or current forest area. This spatial concentration is a target for structural measurement, rather than an identified local correction factor between different masks. The combined classes 4 and 5 have a different mean, $472.19 \text{ MgC ha}^{-1}$, and must be distinguished from class 5 alone.

For a sampled plot of area A_p in hectares, let $a_j > 0$ be a consistently measured stem basal cross-section in m^2 , $H_j > 0$ its height in metres, and $V_j \geq 0$ the reconstructed solid volume of its assigned woody compartments in m^3 . Stem and branch compartments must be disjoint; cavities and external envelope space are excluded from solid material. Crown-boundary and multistem assignment conventions must be fixed. Define

$$\psi_j = \frac{V_j}{a_j H_j}, \quad G_{b,p} = \frac{\sum_j a_j}{A_p} = \nu_p \bar{a}_p, \quad \Lambda_p = \frac{\sum_j a_j H_j \psi_j}{\sum_j a_j}, \quad (\text{S1})$$

where $\nu_p = n_p/A_p$ is stems per hectare. Λ_p is a basal-area-weighted height–architecture quantity in

metres. The dimensionless ψ_j is calculated from independent geometry; it is not a fitted biomass allometry or necessarily a stem-only form factor bounded by one. Branch volume can change its interpretation. Different stem-size distributions may yield the same basal area, so the inventory must retain individual or size-class measurements as well as the aggregate.

The height and architectural contributions separate exactly as

$$\bar{H}_a = \frac{\sum_j a_j H_j}{\sum_j a_j}, \quad \bar{\psi}_{aH} = \frac{\sum_j a_j H_j \psi_j}{\sum_j a_j H_j}, \quad \Lambda = \bar{H}_a \bar{\psi}_{aH}.$$

The first mean is basal-area weighted; the second uses basal-area–height weights. An unweighted mean height cannot be substituted without additional assumptions. At $\bar{H}_a = 30$ m, $G_b = 30$ or 40 m² ha^{−1} would require $\bar{\psi}_{aH} \approx 2.1$ or 1.6 for the ratio-based class-5 target under the illustrative zero-remainder properties. These are ratios of reconstructed solid woody volume to the summed basal-area–height cylinders, not canopy-envelope occupancy or observed forest parameters. Stem taper, branch volume, buttresses, height and cavity treatment must therefore be resolved independently.

Let $\beta_p = \sum_{j,o \in W} V_{jo} \rho_{jo}^b f_{jo}^C / \sum_{j,o \in W} V_{jo}$ be the measured, volume-weighted carbon density of woody material in MgC m^{−3}. Let $c_{\text{rem},p}$ contain the carbon per hectare in all explicitly excluded aboveground compartments, such as separately measured foliage. Bark belongs either in W with matched properties or in the remainder, never in both. Assume total woody volume is positive.

Measurement definitions and carbon reconstruction. For these matched measurements,

$$\frac{V_{Wp}}{A_p} = G_{b,p} \Lambda_p, \quad \boxed{\bar{c}_p = \beta_p G_{b,p} \Lambda_p + c_{\text{rem},p}} \quad (\text{S2})$$

The volume equality follows directly from the definition of ψ_j ; the carbon equality is the sum of material carbon in disjoint compartments. The procedure is therefore tested against independent observations and the mapped prediction, rather than by treating this internal identity as a new theorem. Geometry and material properties must be measured without fitting ψ or β to the target. Held-out plots, fixed compartment rules and shared measurement uncertainty are part of the test.

Table S1: Conditional physical requirements from the published class-5 lower interval edge, both mean-density conventions and the combined classes. The calculation assumes uniform $f_C = 0.47$, basic density $\rho^b = 0.60$ Mg m^{−3} and zero remainder. Biomass is in Mg ha^{−1}, volume in m³ ha^{−1} and equivalent-layer depth in cm. Material properties are illustrative, not measured class means. Approximate biomass and volume are rounded to tens and layer depth to one decimal; full calculations are archived. The lower edge constrains mapped-class cells only if the published interval implements the raster classification; it is not a measured lower bound on every plot or subpixel patch.

Domain	MgC ha ^{−1}	Dry biomass	Woody volume	Layer depth
Class 5: lower class edge	478	~1,020	~1,700	~17.0
Class 5: printed mean	531.05	~1,130	~1,880	~18.8
Class 5: stock / area	536.33	~1,140	~1,900	~19.0
Classes 4 and 5	472.19	~1,000	~1,670	~16.7

Under the illustrative properties in Table S1, $\beta = 0.282$ MgC m^{−3}. The two class-5 mean conventions require approximately 1,880 and 1,900 m³ ha^{−1}, respectively; the approximately one-percent normalization difference leaves the material scale nearly unchanged. The published class interval is 478–605 MgC ha^{−1} [1, Table 3]. At its lower edge, the same properties imply about 1,020 Mg dry biomass and 1,700 m³ of woody material per hectare. If this interval determines raster membership, those lower-edge requirements apply to the *mapped cell estimates* in class 5, not merely their mean. A mapped cell may contain lower-density patches, and an operational map threshold is not an independently observed biological minimum. Verification of the implemented threshold requires the classified raster and area convention.

Stem form, crown allocation and a joint structural test

Let $K = \sum_j a_j H_j$ and partition solid woody volume into disjoint main-axis and branch components, V_s and V_b , with the stump and buttresses assigned explicitly. Define $f_s = V_s/K$, $q_V = V_b/(V_s + V_b)$ and $\psi_W = (V_s + V_b)/K$. Then

$$\psi_W = \frac{f_s}{1 - q_V}.$$

If $q_M = B_b/(B_s + B_b)$ is the branch fraction of dry woody mass and $\kappa = \bar{\rho}_s^b/\bar{\rho}_b^b$ uses the corresponding volume-weighted densities, the same relation is

$$\psi_W = f_s \left(1 + \kappa \frac{q_M}{1 - q_M} \right), \quad \frac{V_b}{V_s} = \frac{\psi_W}{f_s} - 1.$$

A stem form ratio defined with merchantable or bole height H_s must first be converted: $V_s/(aH) = [V_s/(aH_s)](H_s/H)$. A crown envelope is not solid wood. Reported crown mass can include leaves, fruit and differing stem assignments, so it is not automatically q_M . Goodman et al. measured wood density for stems only and defined their crown mass as branches plus attached foliage and fruit; their treewise crown fractions range from 14 to 71%, with mean $44\% \pm 2\%$ [4, pp. 683–685]. Their deliberately size-stratified 51-tree sample is not a stand census. The 673-tree synthesis of Ploton et al., which overlaps earlier datasets, reports approximately 3–88% crown mass [5]. Neither mean is a cylinder-weighted woody-volume ceiling.

Table S2: Structural sensitivity configurations for $c_5 = 536.33 \text{ MgC ha}^{-1}$. Basic density 0.60, carbon fraction 0.47, zero remainder and equal stem/branch densities are specified assumptions. The 0.4–0.6 stem ratio is an illustrative range, not a recovered universal bound. The 0.44 branch fraction approximates a published crown mean only for this simplified two-wood-component calculation. Neither input range is an observed class-5 distribution. $K/A = G_b \bar{H}_a$ is in $\text{m}^3 \text{ ha}^{-1}$.

G_b	$\bar{H}_a$	K/A	Required ψ_W	Carbon at $f_s = 0.4\text{--}0.6$ (MgC ha^{-1})
30	30	900	2.11	181–272
40	30	1,200	1.58	242–363
50	40	2,000	0.95	403–604

For the first two configurations, the required branch-to-main-axis volume ratio ranges from 1.64 to 4.28 as f_s varies from 0.6 to 0.4. Under equal densities this corresponds to branch fractions of approximately 62–81% of woody mass. It is therefore not a fixed three-to-four ratio. For the third configuration, $\psi_W \approx 0.95$ and $q_V = 0.44$ imply $f_s \approx 0.53$, inside the assumed stem-ratio range. Compatibility of a parameter configuration establishes neither its frequency nor its spatial representativeness.

The average of treewise crown fractions cannot be inserted into a plot identity as a bound. With $\alpha_j = a_j H_j / \sum_k a_k H_k$, the plot quantity is

$$\psi_{wp} = \sum_j \alpha_j \frac{f_{s,j}}{1 - q_{V,j}},$$

rather than a ratio formed from unrelated sample means. Two equal-cylinder-weight trees with $f_s = 0.5$ and $q_V = 0.20, 0.68$ have mean $q_V = 0.44$ but $\psi_{wp} = 1.09375$, versus 0.89286 from substituting that mean. This synthetic calculation identifies the aggregation issue without asserting those trees occur in class 5.

A physical rejection is possible without a matched residual if independently established bounds cover the target population. For example, $\beta_p \leq \beta_{\max}$, $G_{b,p} \bar{H}_{a,p} \leq K_{\max}$, $f_{s,j} \leq f_{\max}$, $q_{V,j} \leq q_{\max} < 1$ and $c_{\text{rem},p} \leq R_{\max}$ for all relevant units imply

$$\bar{c}_5 \leq \beta_{\max} K_{\max} \frac{f_{\max}}{1 - q_{\max}} + R_{\max}.$$

The bound requires joint applicability and consistent material definitions; sample means, selected-tree extrema or products of independently observed plot maxima do not supply it. For the illustrative $K_{\max} = 2000$, $\beta_{\max} = 0.282$, $f_{\max} = 0.6$, $q_{\max} = 0.44$ and $R_{\max} = 0$, the bound is about 604 MgC ha⁻¹, above the class mean. At $K_{\max} = 1200$ it is about 363, below it. The difference is precisely the structural premise to establish. The cited TLS studies provide reconstruction checks on selected trees, including modelling of buttresses and small-branch adjustments, rather than a universal plot-level ψ_W maximum [6, 7, Sections 2.3–2.4]. An independent upper envelope below the target would contradict it; the present literature comparison identifies demanding conditions but does not establish that envelope.

For spatial strata s within class 5, with independently specified area weights $\omega_s \geq 0$ summing to one, a class-level reconstruction is

$$\widehat{c}_{5,\text{geom}} = \sum_s \omega_s \overline{\beta_p G_{b,p} \Lambda_p + c_{\text{rem},p}}_s, \quad R_5 = \widehat{c}_{5,\text{geom}} - c_{5,\text{map}}. \quad (\text{S3})$$

The bar denotes the design-weighted mean of plot-level products within a stratum. Multiplying separate means of β , G_b and Λ loses their dependence and generally gives a different answer. The published class ratio is used for $c_{5,\text{map}}$ only on matched class, epoch, pool and area definitions. A representative class mean is the target; an exceptional hectare is not a substitute for the distribution across 531 million hectares.

A forward evaluation reports stem abundance, basal area, the joint height–architecture distribution, sound material volume, density and carbon fractions, remainder pools, inclusion probabilities and propagated errors. For an independent measured-minus-mapped interval $[R_-, R_+]$ and a prespecified margin $\epsilon \geq 0$, $R_+ < -\epsilon$ supports an overestimated mapped target, $R_- > \epsilon$ supports an underestimated target, and $[R_-, R_+] \subseteq [-\epsilon, \epsilon]$ supports agreement at that precision. An interval meeting none of these criteria is unresolved. Attribution to mapping rather than measurement bias additionally requires the independent reconstruction and its uncertainty to be adequately validated. The interval propagates field, calibration and mapping uncertainty and their dependence; the practical equivalence margin is specified independently before comparison. The archive supplies a measured-record schema and a forward evaluator without populating them with purported class-5 observations. Its available source data support the numerical requirements in Table S1; a class-5 structural survey and its original geometry records have not been recovered for an empirical closure result.

S1.1 Comparisons of biomass and remainder-only scenarios

At an explicitly assumed carbon fraction of 0.47, the printed class-5 mean and the ratio-based value imply approximately 1,130 and 1,140 Mg dry biomass per hectare. Against the cited 395.7 Mg per hectare African network estimate, the ratios are approximately 2.86 and 2.88; the 200–350 Mg per hectare Amazonian range gives different comparison factors [8, 9]. These are contrasts between differently supported estimators, not an independently measured upper bound. Basic density changes the volume requirement, whereas $B = C/f_C$ determines the biomass conversion.

For carbon c_5 , positive material carbon density β and a disjoint remainder $c_R < c_5$, the woody-volume requirement is $V_{\text{req}} = (c_5 - c_R)/\beta$. Lowering β raises this requirement; increasing an independently measured remainder lowers it. The archived 65.3% remainder scenario fixes a hypothetical woody comparator at 395.7 Mg per hectare and assigns the residual elsewhere. It therefore tests remainder-only closure under that assumption, not the primary hypothesis that the woody allometric estimate itself is biased. Foliage measurements in the Higuchi et al. sample and living epiphytes in Gómez González et al. do not support such a large nonwoody fraction in those populations [10, 11]; they do not establish a universal class-5 bound. Woody lianas, small stems and already represented branches are not nonwoody residuals. The source component calculations are retained as numerical diagnostics, without converting this restricted scenario into an explanation imposed on the architectural methodology.

An independently reconstructed plot residual can indicate that either mapped estimator is high, low or compatible within a specified margin. Both paths require validated material inputs and common support; failure of one estimator is not inferred solely from agreement of several products with shared calibration labels. Class-level fields in the companion schema remain unpopulated unless independent records are actually supplied.

S1.2 Response-endpoint exceedance and the effect of a numerical cap

Assume the reported class interval is the raster rule on the area being averaged. Let X denote mapped cell carbon under area weights, $a \leq X \leq b$, $\mu = \mathbb{E}[X]$, and $a < T < \mu \leq b$. Then, with $p = \Pr(X > T)$ and $d = \mathbb{E}[(X - T)_+]$,

$$p \geq \frac{\mu - T}{b - T}, \quad \mu - T \leq d \leq \frac{(b - T)(\mu - a)}{b - a}.$$

The first inequality follows from $\mu \leq T(1 - p) + bp$. The lower cap bound follows from convexity of $(x - T)_+$; its graph lies below the chord through $(a, 0)$ and $(b, b - T)$, giving the upper bound. Mixtures on $\{T, b\}$ attain the minimum area fraction and minimum cap change; a mixture on $\{a, b\}$ attains the maximum cap change. A constant $X = \mu > T$ also attains the minimum cap change while placing every cell above the threshold. Thus there is no nontrivial upper bound on affected area from these inputs alone. For area A hectares, $10^{-9}Ad$ is the change in PgC after replacing X by $\min(X, T)$.

Table S3: Conditional class-5 support bounds for $a = 478$, $T = 523$, $b = 605 \text{ MgC ha}^{-1}$ and $A = 531,449,420.52 \text{ ha}$, from the arXiv source [1, pp. 22–23 and Table 3]. The two mean conventions are separate calculations, not endpoints of a statistical interval. Capped-stock changes refer to the selected numerical operation, not demonstrated carbon error.

Mean convention	Minimum area (%)	Area (10^6 ha)	Cap change (PgC)	Max. / 723.97 (%)
Stock-area ratio	16.25	86.4	7.08–20.01	2.76
Printed mean	9.82	52.2	4.28–18.20	2.51

The stock–area mean is $536.3257 \text{ MgC ha}^{-1}$; the alternative 531.05 is the printed mean. Applying the same area to the latter is a normalization sensitivity and does not reproduce the published class stock. All bounds require consistent cell values, area weights and endpoints; no cell-level reconstruction has been recovered. The regional value 523.4 on page 22 yields nearby sensitivity bounds in `data/support_tai1_bounds.csv`. Neither endpoint is a threshold at which true forest carbon is assumed to stop.

The cap can alter less than 3% of the reported total under these conditions while leaving an arbitrarily important calibration discrepancy within the label range. For example, a synthetic map assigning 500 MgC ha^{-1} to every cell has zero change under a 523 cap even if an independent matched measurement is 300. Conversely, a cell above the label range need not be erroneous. Independent labels and joint predictor support, as well as response-range diagnostics, are therefore needed to evaluate magnitude.

For nonnegative compartment carbon C_j with $f_{\min} \leq f_j \leq f_{\max}$,

$$\frac{\sum_j C_j}{f_{\max}} \leq \sum_j \frac{C_j}{f_j} \leq \frac{\sum_j C_j}{f_{\min}}.$$

Using 0.4111 and 0.4999 gives the conditional edge and mean biomass bounds archived in `data/support_tissue_bounds.csv`. The inference assumes those limits cover the mapped material; the observed tissue range alone does not establish universal limits. It requires no assumed basic density because the conversion is carbon to dry mass, not carbon to volume.

S2 Census calibration criteria and derivations

Assume a fixed scalar calibration applies to every event in a matched census. Let j index positive growth intervals $[x_{0j}, x_{1j}]$, $g_j = x_{1j} - x_{0j} > 0$, and d mortality stocks $x_d > 0$. Set $G = \sum_j g_j$, $L = \sum_d x_d$, with weights $v_j = g_j/G$ and $\ell_d = x_d/L$. Recruits may enter with $x_{0j} = 0$ under the specified observed-pool convention. Shrinkage and other losses require additional signed terms, not an unnoticed change in the event set.

Proposition S2.1 (Exact quadratic census criterion). *For $m(x) = \alpha x + kx^2$, with $\alpha > 0$ and $k > 0$ in consistent carbon units, define*

$$\bar{x}_L = \sum_d \ell_d x_d = \frac{\sum_d x_d^2}{L}, \quad \bar{x}_G = \sum_j v_j \frac{x_{0j} + x_{1j}}{2}. \quad (\text{S4})$$

Then

$$r_L - r_G = k(\bar{x}_L - 2\bar{x}_G). \quad (\text{S5})$$

Thus $r_L > r_G$ exactly when $\bar{x}_L > 2\bar{x}_G$; a positive practical margin requires $k(\bar{x}_L - 2\bar{x}_G) > \epsilon_\Delta$.

Proof. For a death, $m(x_d)/x_d = \alpha + kx_d$. For a gain, the exact secant is $\alpha + k(x_{0j} + x_{1j})$. Weighted summation and subtraction prove the result. Neither a small-increment limit nor zero recruitment is assumed. $\square$

The constant scale α cancels from the difference and positive k from its sign, but not from the effect magnitude. The counterexample in Table 8 is exactly $100 < 2 \times 85$. This criterion can screen that calibration family from baseline event data before fitting k ; it is not a calibration-free test of all biological models.

For the power family $m(x) = ax^b$, $a > 0$, $b > 1$ (with the units of a chosen consistently), a finite-interval extension avoids relying on infinitesimal growth. Put $p = b - 1$, and define a probability distribution γ along the gain intervals by

$$\mathbb{E}_\gamma[h(X)] = \frac{1}{G} \sum_j \int_{x_{0j}}^{x_{1j}} h(x) dx, \quad M_{p,L} = (\mathbb{E}_\ell[X^p])^{1/p}, \quad M_{p,G} = (\mathbb{E}_\gamma[X^p])^{1/p}. \quad (\text{S6})$$

Overlapping intervals represent distinct tree events and are counted separately.

Proposition S2.2 (Exact finite-interval power-law criterion). *For these event distributions,*

$$r_L = aM_{p,L}^p, \quad r_G = abM_{p,G}^p, \quad r_L > r_G \Leftrightarrow \frac{M_{p,L}}{M_{p,G}} > b^{1/(b-1)}. \quad (\text{S7})$$

Proof. The loss formula follows by weighting ax_d^{b-1} . The fundamental theorem of calculus gives $x_1^b - x_0^b = b \int_{x_0}^{x_1} x^{b-1} dx$, proving the gain formula. Positive division and the $1/p$ power preserve the inequality. $\square$

The threshold decreases from e as $b \downarrow 1$ to 2 at $b = 2$. Indeed, $\log[b^{1/(b-1)}] = \log(1+p)/p$ decreases for $p > 0$. At $b = 1$, exactly, both factors are a and their difference is zero; the limiting threshold is not a strict-direction test at that endpoint. In the small-increment limit, γ reduces to the event-gain-weighted size distribution. Its p -power mean, not an arithmetic mean substituted at every exponent, remains essential. At $b = 2$, $M_{1,G} = \bar{x}_G$ and the threshold is exactly the quadratic one.

A single arithmetic ratio below 2 does *not* exclude all $1 < b \leq 2$ in heterogeneous censuses. For example, let nine survivors grow from 1 to 1.01, one from 500 to 500.01, and one stock-100 tree die. The

ratio $\bar{x}_L/\bar{x}_G = 100/50.905 \simeq 1.9644$ fails the quadratic screen. Nevertheless, with $m(x) = x^{3/2}$ the exact factors are $r_L = 10$ and $r_G \simeq 4.70749$, so losses are corrected more. This is a synthetic counterexample, not a plausible-sample assertion. The valid screening statistic is exponent-specific: compute the left side of (S7) for the prespecified family. Excluding an entire exponent interval requires the inequality to fail throughout that interval, with numerical tolerances controlled; a few sampled exponents do not prove uniform exclusion. Neither sign screen supplies the calibrated magnitude or independent field support required for H_Δ .

Near-linear calibration. A dimensionally explicit power calibration is $m_b(x) = ax_*(x/x_*)^b$, where $x_* > 0$ is fixed. For $b = 1 + \varepsilon$ and finite logarithmic moments under the loss and gain distributions,

$$r_L - r_G = a\varepsilon [\mathbb{E}_\ell \ln(X/x_*) - \mathbb{E}_\gamma \ln(X/x_*) - 1] + O(\varepsilon^2). \quad (\text{S8})$$

This follows by expanding $(X/x_*)^\varepsilon$ and $(1 + \varepsilon)(X/x_*)^\varepsilon$ in the exact loss and interval-gain operators. Recruitment intervals starting at zero are admissible because $\int_0^h |\ln x|^k dx$ is finite for fixed $h > 0$ and finite k . The differential tends to zero linearly with $b - 1$. Its first-order sign is set by the ratio of loss- and gain-weighted geometric means relative to e ; the reference stock cancels in their log difference. If the coefficient vanishes, higher-order terms determine the sign.

S3 Historical emissions-percentage reconstruction

S3.1 Tracing the 10.3% benchmark

The historical preprint's emissions discussion cites Ecofys's *World GHG Emissions Flow Chart 2010*, issued in 2013 [1, 12]. The original chart states a global total of 48,629 MtCO₂e, with all percentages referring to that total. The 10.3% label belongs to *deforestation / afforestation and land-use* CO₂. It is therefore a combined land-use category, not a separately identified gross tropical-deforestation flow.

Its displayed share implies approximately $0.103 \times 48.629 \simeq 5.01$ GtCO₂ for that CO₂-labelled component. This is a calculation from rounded graphical inputs, not precision established by independent data. Because the numerator is CO₂, using the multi-gas denominator is dimensionally permissible; the substantive problem is category and source compatibility, not the mere presence of CO₂ inside a CO₂e chart.

The chart cites IEA, EDGAR, CDIAC and Global Carbon Project source vintages. It does not specify a recoverable link from the 10.3% numerator to either the 228.7 or 285.1 PgC stock map, and its plate does not identify a unique GWP assessment basis. The cited plate is the original Ecofys-authored chart preserved by an external mirror. Its original publisher URL and retrieval limitations are recorded in the provenance notes; no unverified Wayback capture is substituted for an accessible source. The exact component assembly and spatial density correspondence therefore remain unverified. These omissions are not filled by assuming that all contemporaneous land-use estimates used the same carbon-density model.

The conversion of a stock contrast into a gross-deforestation percentage requires a common activity definition, spatial basis and stock-to-flux operator. The historical 10.3% category combines land-use effects and removals, so the displayed multiplication is interpreted here as a conditional accounting scenario. These requirements concern the annual inference even if the 723.97 PgC stock is accepted in full. They do not constitute a test that rejects that stock estimate.

S3.2 Two arithmetic reconstructions, not two measurements

Let $s_0 = 0.103$ denote the chart's combined category share, and provisionally assign it a scalar multiplier r_L . For the arithmetic demonstration only, set $r_L = r_C$ using the displayed stock ratios:

$$r_{C,1} = 723.97/285.1, \quad r_{C,2} = 723.97/228.7. \quad (\text{S9})$$

The fixed-total calculation and the retained-other-contributions calculation are respectively

$$s_{\text{fixed}} = r_L s_0, \quad s_{\text{renorm}} = \frac{r_L s_0}{1 + (r_L - 1)s_0}. \quad (\text{S10})$$

The latter follows by adding the component increment once to the total. The former requires an offset elsewhere to retain that total. Neither establishes that the combined historical land-use component has the imposed multiplier.

Table S4: Reconstruction from displayed historical inputs. The original published range is 26.15–32.60%; nearest rounding of the displayed inputs gives 26.16–32.61%, whereas truncation gives the published 26.15–32.60%. No endpoint is a confidence limit. $r_L = r_C$ is imposed despite unverified spatial and numerator compatibility.

Comparison stock (PgC)	Ratio	Fixed total (%)	Retained others (%)
285.1	2.53935	26.16	22.58
228.7	3.16559	32.61	26.66

The renormalized interval is 22.58–26.66%. The historical fixed-total endpoints are compatible with at least two display paths: truncating the calculation with a baseline of exactly 10.3%, or rounding a calculation with an unprinted baseline share of approximately 10.296665–10.299824%, which itself displays as 10.3%. The percentage-column pattern favors truncation as a coherent explanation, but the source calculation has not been recovered to identify its actual path. Both possibilities are recorded rather than assigning the endpoint difference to an established cause. This distinguishes denominator arithmetic from the much harder stock-to-flux inference. For a harmonized net land component, uniform paired scaling could produce a common multiplier of its calibrated net part, as (25) shows. That mathematical possibility does not establish that Ecofys’s component, the historical map totals and the relevant removal estimates meet those conditions.

Figure 2 displays the arXiv-reported range in a separately labelled historical annotation, together with the renormalized component range and the original and reconstructed totals. The question mark and double-headed arrow concern the total and its attribution; they do not identify an observed error in a fossil source. The standing-stock record remains separate from the annual-flow network.

S3.3 Sector ranking under a retained-component account

The numerical sector comparison must use a single grouping and period. Ecofys’s 2010 sector called Land Use Change contains both the 10.3% component examined above and agricultural soils at 4.4%. Its industry outputs total $22.8 + 6.3 = 29.1\%$, rounded to 29% in the sector heading [12]. That land grouping is not equivalent to the modern EDGAR LULUCF boundary. The conceptual WRI-style network does not supply a new numerical sector allocation; the historical card explicitly identifies the separate Ecofys grouping.

Set $a_0 = 0.044$, $i_0 = 0.291$, and $T_0 = 48.629$ GtCO₂e. With the historical multiplier r applied only to $s_0 = 0.103$, retain the rest of the land group and all other absolute contributions. Then

$$T^* = T_0[1 + (r - 1)s_0], \quad L^* = T_0(rs_0 + a_0), \quad I^* = T_0i_0.$$

The other sources and any removals already represented in those terms are retained by construction. In this scenario, $L^* > I^*$ precisely when

$$r > \frac{i_0 - a_0}{s_0} = 2.398058 \dots$$

Both imposed stock-ratio endpoints exceed that threshold. Dividing all sectors by the same $T^* > 0$ preserves their ordering. This is the precise retained-component largest-sector inference, independent of whether the comparison is expressed in annual masses or consistently renormalized percentages.

Table S5: Conditional 2010 sector ranking, calculated from source-displayed inputs. F^*/T_0 uses the original total; L^*/T_0 includes the retained 4.4% agricultural-soils contribution. Final shares both use the same revised total. The historical multipliers are imposed, not empirically estimated annual corrections. Ecofys’s land-use grouping differs from modern LULUCF.

Stock (PgC)	$100F^*/T_0$	$100L^*/T_0$	T^* (GtCO ₂ e)	$100L^*/T^*$	$100I^*/T^*$
285.1	26.16	30.56	56.34	26.37	25.12
228.7	32.61	37.01	59.48	30.26	23.79

For the forest-related component alone, the threshold is instead $r > i_0/s_0 = 2.825243 \dots$: only the upper historical endpoint exceeds it. A sector-ranking conclusion therefore requires its full boundary; it cannot be transferred between a forest component, a broader land-use group, net LULUCF and the energy supersector. A fixed-total scenario additionally requires an offset elsewhere, whereas the retained-component calculation above increases the total to approximately 56.34–59.48 GtCO₂e and keeps industry at approximately 14.15 GtCO₂e. Keeping both the old total and every old component while adding a positive correction is inconsistent.

The original diagram also assembled labels from several reporting years. Numerical rankings here use the 2010 Ecofys allocation rather than compare a 2010 land reconstruction against a power share from another reporting year. Common spatial support, carbon-pool definitions, source provenance and annual transfer still condition empirical application. The capped and recalibrated map calculations do not by themselves estimate those annual operators.

S4 Published source-table transcription and label diagnostics

Table 3 of the preprint reports 723.97 PgC over 2,799,053,606.91 ha within its mapped pantropical domain, excluding the Australian tropics. Its classes include high-carbon forest, disturbed vegetation, woodland, scrub, savanna and degraded land; the denominator is therefore a mixed vegetation domain, not a rainforest-only area [1]. For a stock C in PgC, area A in hectares and assumed dry-mass carbon fraction f_C ,

$$\bar{c} = 10^9 C/A, \quad \bar{B} = \bar{c}/f_C. \quad (\text{S11})$$

Using the source’s displayed totals gives $\bar{c} = 258.65 \text{ MgC ha}^{-1}$. The two highest-carbon classes contain 566.79 PgC over 1,200,345,923.20 ha, implying $472.19 \text{ MgC ha}^{-1}$. Those carbon densities do not depend on any biomass back-conversion.

The original method reports measured tissue-carbon fractions of 0.4111–0.4999 and predicts carbon directly through its geometric and neural-network calibration [1], rather than specifying a uniform pantropical factor of 0.50. Reconstructing the biomass actually implied by that method requires the associated biomass weights: for compartments j , $B_\Sigma = \sum_j C_j/f_j$ and $f_{\text{eff}} = C_\Sigma/B_\Sigma$. The published aggregate does not recover these weights or an effective pantropical fraction. Neither inversion by a tissue-average fraction nor division by 0.50 reproduces the original biomass in their absence.

As explicit sensitivities, $f_C = 0.47$ gives $550.32 \text{ Mg dry biomass ha}^{-1}$ for the mixed domain and 1004.66 for the high-carbon classes; $f_C = 0.50$ gives 517.30 and 944.38 , respectively. If all mapped compartments additionally had fractions within the measured 0.4111–0.4999 range, the corresponding means would lie within 517.40 – 629.16 and 944.57 – 1148.60 . These are conditional back-conversions, not a recovered biomass surface. In particular, the approximately 1005 Mg ha^{-1} value follows from an assumed 0.47 fraction; it is neither a mandatory output of the original carbon method nor evidence of a physical impossibility. Carbon mass per hectare and the weight of an individual tree are different quantities.

The two intermediate numerical rows in the published table are mislabelled relative to its density-class

Table S6: Published-digit audit of the historical source’s Table 3 [1]. Intervals, printed means and percentages are transcribed; the ratio column is $10^9 C/A$. Carbon-density units are MgC ha^{-1} . Areas are shown in million hectares here, with the full two-decimal-hectare values in the repository. Classes 2 and 3 fit each other’s intervals. The printed total is 723.97 PgC.

Class	Printed interval	Area (10^6 ha)	PgC	Printed mean	Ratio	Printed %
5	478–605	531.449	285.03	531.05	536.33	39.37
4	336–478	668.897	281.76	417.09	421.23	38.91
2	58–183	419.412	105.54	249.16	251.64	14.57
3	183–336	350.689	40.66	114.80	115.94	5.61
1	1–58	828.607	10.96	13.10	13.23	1.51

definitions. The row with stock–area ratio $251.64 \text{ MgC ha}^{-1}$ belongs to class 3 (183–336), and the row with ratio 115.94 belongs to class 2 (58–183). Their printed means give the same assignment. Because each numerical pair lies strictly inside one of the existing intervals, direct membership fixes the assignment; in source-row order it is (5, 4, 3, 2, 1) rather than (5, 4, 2, 3, 1). This corrects the printed numerical-row/class association and restores increasing density with class while retaining every stock and area. The source transcription remains separate from the corrected assignment. The operational key is needed to determine whether and where the original vegetation descriptions or raster legend share the labelling problem; the table correction does not establish an error in pixel carbon. Enumeration of the 120 assignments is retained only as a regression check.

The mean-density discrepancy has a common pattern. Define $q_j = \bar{c}_{j,\text{printed}}/(10^9 C_j/A_j)$. The five values in source order are approximately 0.990163, 0.990169, 0.990153, 0.990139 and 0.990397. Under nearest rounding, a value printed to two decimals represents an interval with half-width 0.005; under positive-number truncation, it represents the interval from the printed value to that value plus 0.01. Applying the respective interval rules to stock, area and mean gives common-factor bounds [0.990140, 0.990190] and [0.990134, 0.990182]. Both models therefore accommodate one shared normalization. Exact division by 1.01 is excluded by classes 4 and 5 under either rule; the available digits identify an approximately one-percent convention difference, rather than its computational origin.

All five numerical rows are compatible, within displayed precision, with one common multiplicative normalization; separate class-specific factors are unnecessary. The representative $q = 0.99016$ gives $1/q = 1.0099378$. Under a shared carbon numerator, $q = A_{\text{table}}/A_{\text{mean}}$, so a denominator about 0.994% larger would reproduce the central factor. A common multiplier is a parsimonious description, but its processing stage is not identified. In particular, a spatial area multiplier $s(x) > 0$ gives $q_j = A_j / \int_{\Omega_j} s(x) dA = 1/\mathbb{E}_{A,j}[s]$. Different class footprints can have the same class-weighted mean even when s varies within them. Thus a spatial explanation must reproduce the common class-level factor, rather than being categorically excluded. Class 1’s central-factor residual is only $0.00314 \text{ MgC ha}^{-1}$, below half a displayed mean-density unit. Its weaker relative precision supplies no separate proof of a projection effect. Mean-column operations, pixel-area weights and denominator records distinguish the alternatives.

There is direct evidence for truncation in the percentage column. Truncating $100C_j/723.97$ to two decimals reproduces all five percentages in Table S6; nearest rounding does not reproduce classes 4, 2 and 3. The stock sum is 723.95 PgC against the printed 723.97, and the area sum is 2,799,053,606.89 ha against the printed 2,799,053,606.91. Both discrepancies are two units of the last displayed digit. These results are consistent with truncating component values and a separately computed total.

A probability calculation can compare display conventions only after its assumptions are specified. For five independent uniform hidden component fractions and a total computed from their exact sum under the

same display rule, a positive two-unit gap has probability $11/20 = 0.55$ under truncation and $1/120$ under nearest rounding; the probability of a gap of either sign with magnitude two under nearest rounding is $1/60$. These are conditional reference-model probabilities, not posterior probabilities of the historical workflow. Carbon and area outputs may share computations, so their probabilities are not multiplied. Truncation explains the visible pattern parsimoniously, while the original calculation is needed to identify the actual display convention.

The printed-table correction, common-factor compatibility and historical processing origin are separate results. The first is determined by interval membership and leaves carbon quantities unchanged. The second follows from the digit intervals. The third requires the original workbook, class key and area computations, which were not recovered among the available records. The source transcription and corrected class association are supplied separately in the numerical archive.

Spatial support determines the meaning of a comparison. Feldpausch et al.'s 285.1 PgC total uses 1667.5 million hectares of specified forest classes, with a mean of $170.97 \text{ MgC ha}^{-1}$ [2]. For any two stock–area totals $C = A\bar{c}$, their ratio decomposes as

$$\frac{C_A}{C_B} = \frac{A_A}{A_B} \frac{\bar{c}_A}{\bar{c}_B}. \quad (\text{S12})$$

The complete historical comparison is approximately $2.539 \approx 1.679 \times 1.513$: both the larger area and the different mean density contribute to the numerical ratio. This is an arithmetic factorization, not a causal attribution of the difference. The broader historical domain includes extensive lower-carbon vegetation, so its mixed-domain mean attenuates the high-class density contrast.

Classes 4 and 5 contain 566.79 PgC on 1200.346 million hectares: 1.988 times Feldpausch's entire stock on 0.720 times its area. Their mean density is 2.762 times the whole-domain comparison mean. This highlights the high-density requirement but is not yet a matched-pixel correction factor. Let H denote those classes and $C_{0,H}$ their conventional carbon. If H is contained in Feldpausch's mask, then $0 < C_{0,H} \leq 285.1$ and

$$r_H = \frac{566.79}{C_{0,H}} \geq \frac{566.79}{285.1} \approx 1.99. \quad (\text{S13})$$

The value 2.76 is obtained if H had the conventional whole-domain mean density. If its conventional mean is additionally at least that whole-domain mean, the conditional bracket becomes $1.99 \lesssim r_H \lesssim 2.76$. Without this extra density assumption, 2.76 is a comparison scenario, not an upper bound: a lower conventional local density can require a larger factor. Neither nesting nor that density ordering has been established by raster intersection here. The repository supplies explicit counterexamples to an unconditional finite bracket.

A common-mask analysis is needed to estimate $C_{0,H}$ and separate biological calibration from vegetation selection. The comparison with 228.7 PgC also requires matched pools, vegetation support and period [13]; no associated Baccini area has been recovered for this audit. Equation (S12) is therefore evaluated for Feldpausch only, while the Baccini historical endpoint remains a total-stock arithmetic reconstruction.

S5 Compact statistical and methodological notes

S5.1 Reconciliation and structural identifiability

For an observation model $y = Hx + \epsilon$ with parameter covariance or likelihood specified independently, a linear combination $a^T x$ is identifiable from noiseless Hx exactly when a belongs to the row space of H . To see this, two parameter vectors with the same observation differ by $v \in \ker H$. Their target values agree precisely when $a^T v = 0$ for every such v , equivalently $a \in (\ker H)^\perp$. This is the general form of the source–sink ambiguity in (31).

Under an independently specified Gaussian reference x_0 with covariance Σ and an observation-error covariance V , the unconstrained conditional mean is

$$\hat{x} = x_0 + \Sigma H^T (H \Sigma H^T + V)^{-1} (y - H x_0). \quad (\text{S14})$$

This standard result does not add information absent from the covariance and observations. Hard physical bounds or non-Gaussian discrepancies require an appropriate constrained or non-linear analysis. A statistically unique posterior mean produced by a prior is not the same as structural identification by the data. Shared allometric calibration belongs in covariance or explicit model-discrepancy terms, not in a repeated claim of independent evidence.

S5.2 Bias, covariance and ratios

For a squared-error estimator, $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$. Additional samples can reduce sampling variance without eliminating a common model error. Inventory uncertainty may involve activity data, emission factors and correlated aggregation [14]. The paired correction is a structured instance of such dependence, not a claim that all uncertainties are equal or symmetric.

For a ratio $s = F/T$ with nonzero T , first-order propagation gives

$$\text{Var}(s) \simeq \frac{\text{Var}(F)}{T^2} + \frac{F^2 \text{Var}(T)}{T^4} - \frac{2F \text{Cov}(F, T)}{T^3}. \quad (\text{S15})$$

If T contains F , independence is generally unjustified. The approximation is unreliable near a zero or poorly constrained denominator. Joint simulation or explicit ratio inference is then preferable. The present article does not fit an uncertainty interval around the historical reconstructed percentages.

S5.3 Model domains and admissibility

A unit with zero baseline carbon and zero corrected contribution may be omitted from the restricted fixed-stock release representation. That permission does not extend to regrowth from zero stock, positive corrected flux on a zero baseline increment, or newly represented carbon pools. Those cases require the increment or additive terms in Section 7.2. Newly included areas are a support change and must be accounted for separately. Positive baseline flux is necessary to define a multiplier. Signed net weights do not satisfy the convex-combination bounds for r_F ; separating positive source and removal components avoids that misuse. The general net correction in (19) does not require common calibration; only its common- r_C expansion does. Separate bases admit two transfer identities, but no common covariance term follows without an explicit mapping. The algebraic identity for a net correction remains valid when its source and removal ratios are supplied by different admissible operators; the common-stock covariance formula does not.

For repeated censuses, additions and removals of sampling units must be reconciled before invoking scalar change identities. For delayed release, the cohort index can absorb spatial position, disturbance year, tree-size compartment, organ and carbon pool. A change in decomposition kernel, combustion completeness or regrowth trajectory is a process correction, not merely a carbon-stock multiplier. These distinctions are necessary both for causal interpretation and for conservation of carbon in an implementation.

S6 Published-data application and residual identifiability

The input audit uses the *Acalypha* sp.–*Guazuma ulmifolia* community in Vásquez and Arellano [15, Table 170, p. 942]. The published observations are community summaries per 0.05 ha, rather than a new single-plot survey. The source-method carbon combines allometric biomass and measured tissue fractions. Those quantities are retained under their actual measurement status.

Table S7: Worked audit of real published Córdoba aggregates. Per-hectare results divide source totals by 0.05 ha. Class height ranges bound the basal-area–height cylinder sum from displayed inputs; the sum is not solid woody volume. Minor aggregate differences follow displayed precision.

Quantity	Reproduced result
Reported mean stem density	1,593.4 ha ⁻¹
Reported mean basal area	45.2 m ² ha ⁻¹
Sum of displayed class basal areas	45.4 m ² ha ⁻¹
Allometry-based aboveground biomass	147.8 Mg ha ⁻¹
Source-method carbon	66.8 MgC ha ⁻¹
Displayed-class $\sum_h G_h H_h$ envelope	236.5–778.4 m ³ ha ⁻¹
Independent geometric carbon / mapped residual	Not identifiable

With reported class basal areas G_h and height limits $H_{h,-}$, $H_{h,+}$, the diagnostic is

$$\sum_h G_h H_{h,-} \leq \sum_j a_j H_j / A \leq \sum_h G_h H_{h,+}.$$

Applied to displayed class aggregates it gives a nominal cylinder-sum envelope, not an error interval or a woody-volume estimate. It does not supply the unknown joint distribution of ψ , density and chemistry. The sum of the rounded class basal areas is 45.4 m² ha⁻¹, versus the separately printed total 45.2; neither is silently substituted for the other.

The source summary lacks independent sound organ volumes, their paired material properties, and a recovered map value and class attached to the same georeferenced footprint and epoch. Converting its allometric biomass back into volume would recycle the estimator under test; assigning a mapped class from the source-method carbon would assume the desired map correspondence. Accordingly, `data/cordoba_empirical_audit.json` records the reproduced quantities, absent inputs and a null geometric residual. The executable eligibility check thus prevents an unsupported empirical claim. The separately reported water-displacement comparison tests reconstructed volume at its sampled scale and does not furnish the missing class-level carbon residual.

S7 Evidence from harvest, laser scanning and forest networks

The regional contrast reported on arXiv is larger than the approximately 1.1–2.2 upward contrasts discussed for independent architectural, local-calibration and landscape comparisons in the article and this table. These quantities have different denominators and biological supports. Their direction supplies third-party evidence for a mechanism; their magnitudes neither validate 2.88 across the pantropics nor impose an upper bound on the possible bias of an unmeasured population.

Reference methods require the same scrutiny as the allometries under test. Gonzalez de Tanago et al. [7] used nine of 29 harvested trees for reconstruction calibration and 20 for independent validation. Their Equation 3 defines bias by the mean residual divided by mean reference AGB, equivalently

$$b = \frac{\sum_j (\hat{B}_j - B_j^{\text{ref}})}{\sum_j B_j^{\text{ref}}}, \quad \frac{\sum_j B_j^{\text{ref}}}{\sum_j \hat{B}_j} = \frac{1}{1 + b}. \quad (\text{S16})$$

Thus $1/(1 - d)$ is valid for a reported negative bias $b = -d$, but as a ratio of *sample totals*, not the mean of individual replacement factors. Their Table 4 explicitly identifies 20 validation trees and gives endpoint factors of approximately 1.180–1.554. Sampling selected the largest tree in each of 29 plots (mean diameter 73.5 cm across the sampled trees). The reported increase of allometric underestimation with size cautions against extrapolating this large-tree sample to a stand: a smaller-tree-weighted population would

Table S8: Selected primary evidence and the limit of its transfer to a global correction. Percentages have different denominators and must not be pooled as interchangeable effect sizes. Dry-mass densities are in Mg ha^{-1} .

Study / support	Reported result	Interpretation limit
Arellano-P. and Rangel-Ch. (2015/2016), arXiv source; 54 regional vegetation types	Reported mean carbon 177.03 versus 61.41 MgC ha^{-1} ; ratio 2.88.	Author's regional method contrast, not an independent or pantropical validation factor; the ratio of displayed means is distinct from any area-weighted global correction.
Gonzalez de Tanago et al. (2018); 29 trees, of which 20 formed the independent validation set	Harvest-geometry reference: TLS bias -3.68% ; pantropical-model biases -15.22 to -35.67% .	Geometry and assigned density, not complete mass weighing. Aggregate validation-sample replacement factors approximately 1.180–1.554.
Momo Takoudjou et al. (2018); 61 harvested trees, Cameroon	Edited quantitative structure models had 4.68% volume bias; TLS- and harvest-calibrated allometric coefficients were statistically indistinguishable.	The quoted percentage is <i>volume</i> , not global carbon bias. Manual model editing and density remain consequential.
Huertas et al. (2025); 118.75 ha of French Guianan plots	Local species-specific ALS height–diameter models produced 11–13% higher AGB estimates.	Local model comparison; not independent weighing of a pantropical carbon stock.
Demol et al. (2024); miombo landscape	Carbon estimates 1.5–2.2 times conventional comparison estimates at landscape scale.	Dry woodland and scaling methods differ from moist-forest and whole-tropics inference.
Kauffman et al. (1995); four slashed Amazonian primary forests	Total aboveground fuel biomass 292–436; forest-floor material is included.	Clearing-site inventories are not equivalent to complete, independent weighing of all live trees in one hectare. Pools differ from live AGB.
Malhi et al. (2006); Amazonian plot-network synthesis	Regional aboveground live biomass approximately 200–350, with spatial variation and additional small-stem/liana accounting.	Allometry-dependent spatial estimates, useful as contextual comparison, not an independent absolute ceiling.
Lewis et al. (2013); 260 intact African plots	Mean AGB 395.7; 95% confidence interval on the mean ± 14.3 .	Intact closed-canopy sample, not all vegetation; reference calibration can be shared.
Feldpausch et al. (2012); height-integrated tropical analysis	Including height reduced the estimated total from 320.5 to 285.1 PgC over the same forest mask.	A counter-direction model result; compared masks must remain identical.
Avitabile et al. (2016); fused pantropical map	Fusion and bias adjustment gave a lower total than the input-map estimates.	Independent map differences can involve support, calibration and fusion, not one universal biological multiplier.

have a lower aggregate factor only if the same size-dependent calibration and relevant weighting also apply. The reference AGB itself was calculated from harvested-tree geometric measurements, assigned wood density and small-branch expansion, not by weighing all reference biomass; the authors explicitly acknowledge those limitations. This is informative structural validation, but not independent validation of every density or chemical input. Momo Takoudjou et al. [6] likewise demonstrate the importance of checking reconstruction quality rather than declaring laser-derived models error-free.

The moist-forest local-calibration and miombo findings demonstrate that substantive differences can arise from architecture, species calibration and mapping scale [16, 17]. The mixed-domain density contrast of approximately 1.51 lies numerically within the Gonzalez de Tanago and miombo ranges. Under nested masks, the high-class comparison gives a lower factor near 1.99; the 2.76 scenario additionally assigns a whole-domain density to that subset. These comparisons describe different populations, supports and estimators. A regional factor is neither confirmation of a pantropical total nor an upper bound on the correction permitted elsewhere. Height integration and map fusion can also move estimates downward [2, 18], emphasizing that each result must be interpreted through its calibration chain.

The network contrasts are quantified in Supplementary Note S1: class-5 conditional biomass is 2.88 times the cited African mean and 3.26–5.71 times the Amazonian regional range [8, 9]. Their allometric dependence and different supports require an independent class-level material comparison, not omission of the gap. Kauffman et al.’s 292–436 Mg ha⁻¹ clearing-site range refers to aboveground fuel pools [19]; it must be distinguished from the live-biomass network quantities. The forward protocol tests the relevant carbon-volume relation with independent geometry, chemistry, compartment completeness and spatial selection.

A genuine whole-hectare destructive mass census would strengthen this comparison, but a hectare-scale clearing experiment must not be relabelled as such solely because its footprint is one hectare. Inventory-derived initial masses and sampled burn consumption remain model-assisted estimates. Here harvested-tree geometry comparisons, any directly weighed reference material, clearing inventories and plot networks remain distinct evidential classes. This distinction prevents an apparently independent benchmark from covertly reusing the calibration being tested.

S8 Reproducibility and source provenance

S8.1 Inputs, outputs and numerical checks

The companion archive contains the exact displayed source entries in `data/inputs.json`, including class order, full-precision areas, means, percentages and intervals. It also records the Ecofys category and denominator, the Gonzalez de Tanago validation metric and explicitly synthetic examples. The published table is left unchanged by the diagnostic calculations.

`scripts/reproduce.py`, `scripts/extensions.py` and `scripts/audit_extensions.py` regenerate the numerical and LaTeX-table outputs listed in the manifest. These cover density and support comparisons, crossed intervals, nearest-rounding and truncation models, exact conditional gap probabilities, paired accounts, zero-stock regrowth, finite-increment census criteria, single- and mixed-cohort release, near-linear calibration and invalid inputs. Machine-readable values and typeset tables share the same calculations.

Exact suite counts, random seeds and tolerances are recorded in the README and validation JSON files; these computational checks assess stated algebra and transcriptions, not independent class-5 measurements.

The `stand_measurement_schema.json` and forward evaluator retain measured-record provenance, independent compartments and spatial weights. The capture-route schema records the additional acquisition metadata needed for the initial reconstruction and future calibrations without inventing historical tree records. All generated values identify whether they are source transcriptions, synthetic tests or conditional

computations.

S8.2 Archive and repository verification

The repository manifest lists relative paths, byte counts and SHA-256 digests for the research snapshot. The verifier checks integrity and compares regenerated numerical outputs within stated cross-platform floating-point tolerances; non-numerical values and CSV/LaTeX text remain exact comparisons. Build reports identify compiler versions, warnings and PDF preflight results. These are integrity and computational-reproduction checks, rather than digital-authenticity certification or biological validation. Source-observation years, publication years and the research-snapshot date are recorded separately.

S8.3 Empirical scope and unrecovered records

The numerical archive reconstructs published tables and tests exact identities. It includes a source-transcribed intermediate-stem example, near-100 kg architectural records and deterministic atmospheric-account demonstrations; source-reported shares remain distinct from derived means. A common-mask carbon integration, original classification-pipeline rerun, reconstructed Ecofys component inventory and matched field census analysis require additional empirical records. The source-availability note identifies the workbook, raster, class-key and carbon-fraction weights that were not recovered, without assigning an unverified reason for their absence. The table-level interval assignment is unique under the stated fixed scheme. Its translation to a historical raster legend, and the computational cause of the common normalization, require the original records.

S8.4 Calibration-support and process-account provenance

The source record separates arXiv dissemination (2015/2016), the nominal 2007–2008 map epoch, and the ongoing successor calibration. Page 23 of the preprint supplies the 0–523 MgC ha⁻¹ response range; it does not supply a record-by-record capture manifest or an independent class-5 residual. The Global Carbon Budget inputs retain period, gas, sign and uncertainty definitions. Its detailed 2024 atmospheric section reports ± 0.2 GtC, whereas an executive-summary sentence prints ± 0.02 ; the adopted value and location are explicit in the source record. TLS2trees performance numbers are hardware- and implementation-specific benchmarks. `scripts/review_update.py` recomputes the support comparisons, selected gross-pair sensitivity and illustrative observational tests; `data/review_update_sources.json` records the attribution.

`scripts/support_budget.py` reproduces the class-5 moment bounds and tissue-fraction sensitivities, checks equality cases and randomized bounded distributions, and calculates the disaggregated GCB process pairs and cumulative-residual scenarios. Source values, rounded-input subtraction, selected sensitivity parameters and interval assumptions are distinguished in `data/support_budget_inputs.json`. The cap calculation constrains a numerical operation, not true calibration error. The preprint's 523.4 regional value is retained as an endpoint sensitivity. No missing raster cells, empirical capture records or geographic correction fractions are fabricated.

The additional `scripts/structural_constraints.py` generates five files: the conditional stem/branch scenarios, a typeset table, the weighted-allocation counterexample and joint-envelope calculation, the two-source ranking threshold, and its validation record. The 1,000 synthetic cases use seed 20260925 and contribute 8,017 assertions with the same numerical tolerances. No independent tree-level geometry or class-5 observations are generated. The complete reproduction command yields 76 numerical/table outputs.

The module `scripts/sector_ranking.py` adds the historical retained-component sector calculation and ranking-invariance tests. Source inputs and the distinction between the Ecofys land-use group and modern LULUCF are specified in `data/sector_ranking_inputs.json`. With seed 20260918, 1,000

synthetic positive-source partitions test accounting closure, common-denominator ranking invariance and the analytic threshold. Four generated files preserve the historical example, typeset table, explanatory scope and test result.

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